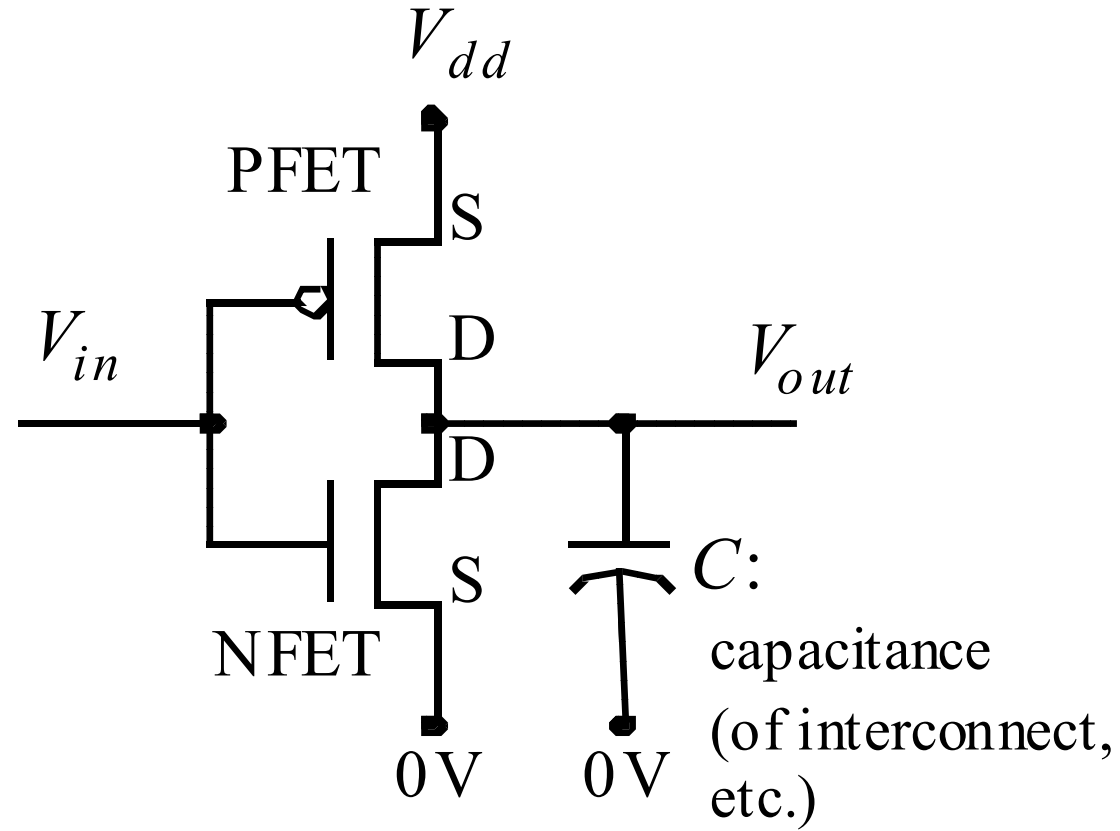


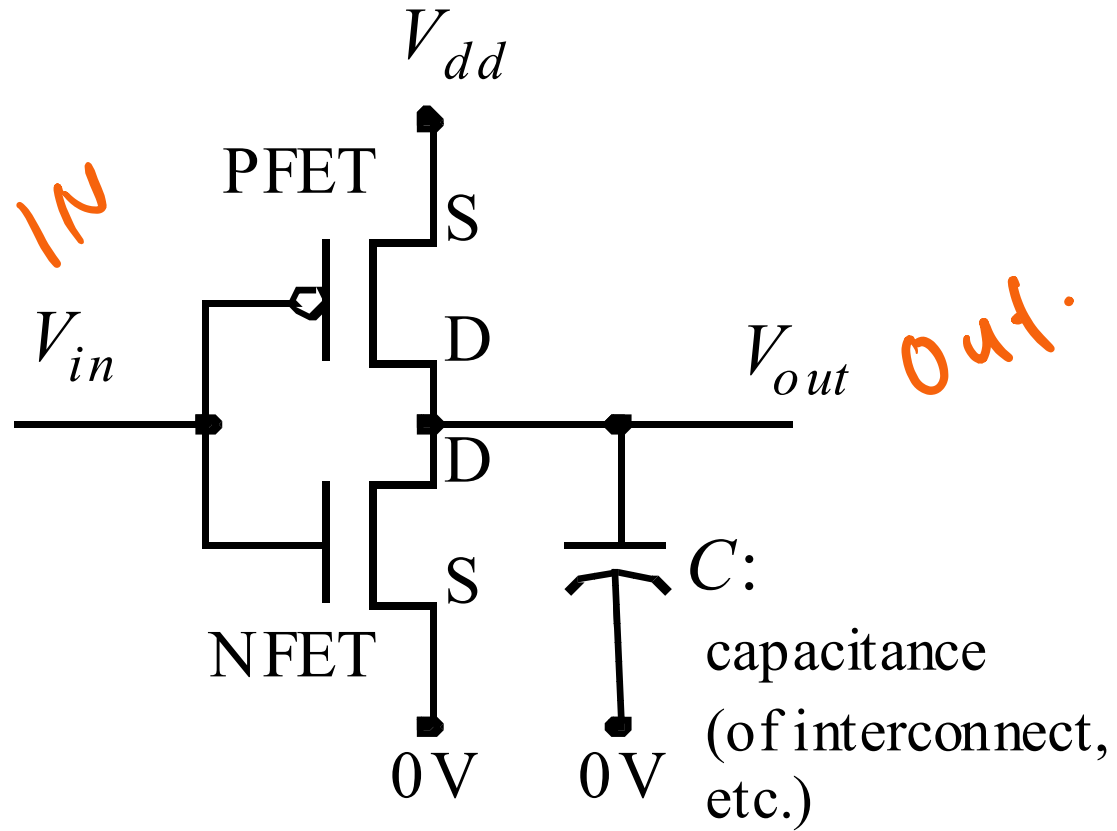
ECE 105: Introduction to Electrical Engineering

Lecture 12
Optoelectronics
Yasser Khan
Rehan Kapadia

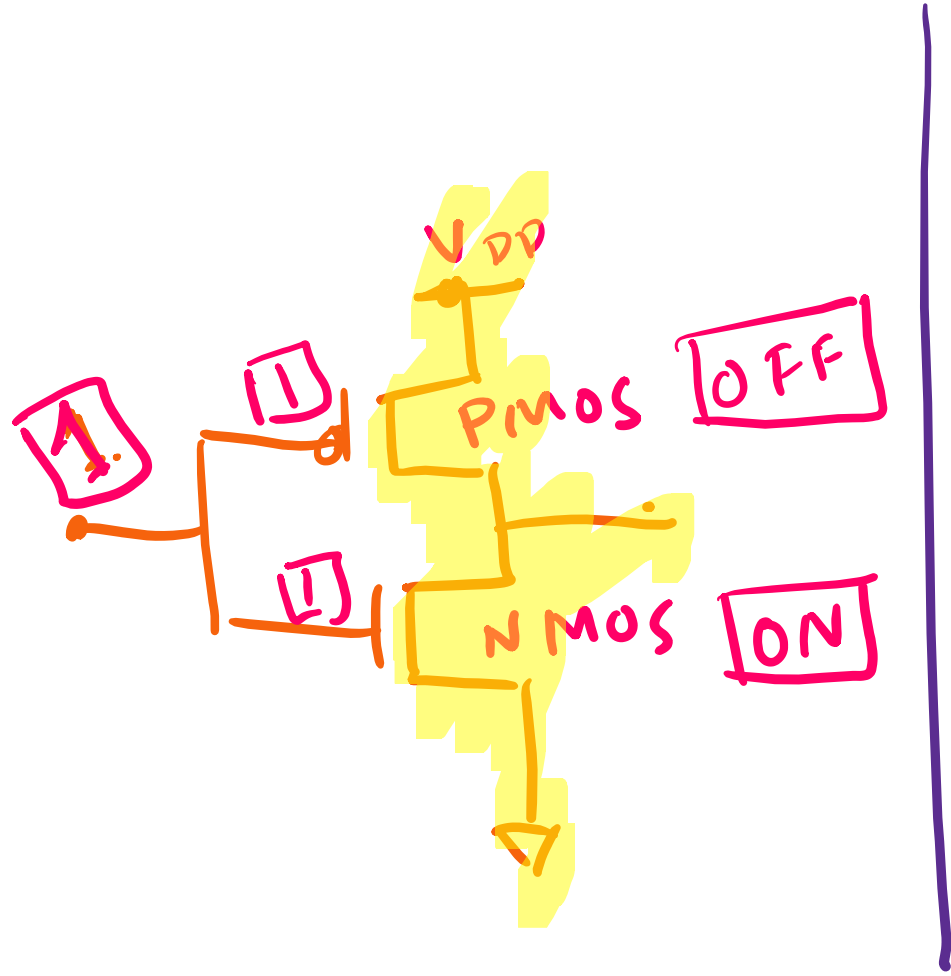
CMOS (Complementary MOS) Inverter



CMOS (Complementary MOS) Inverter



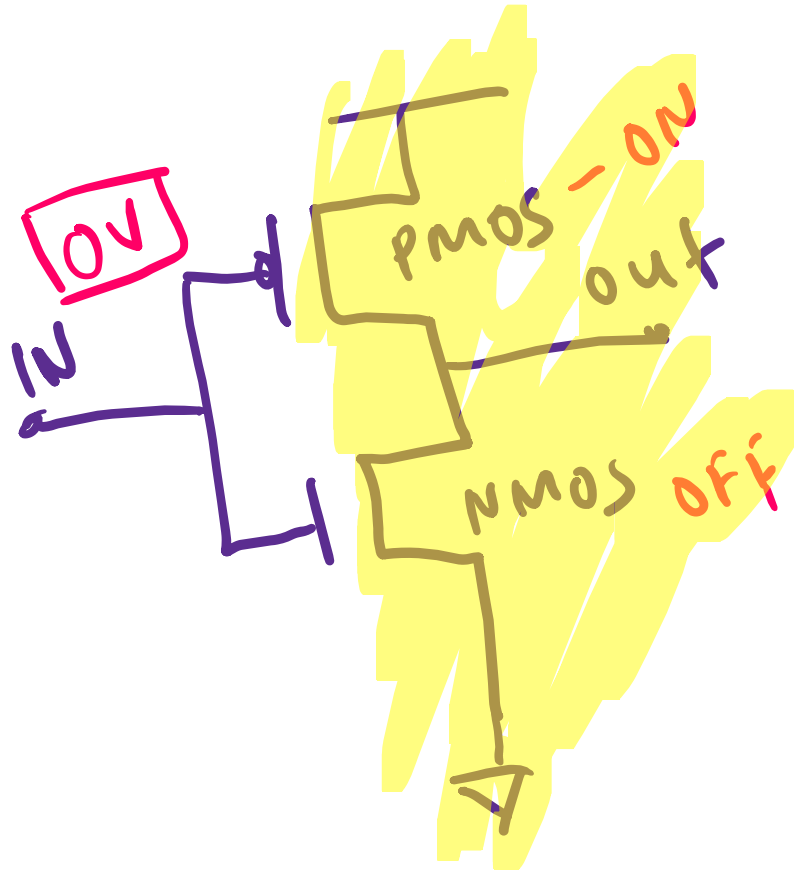
CMOS (Complementary MOS) Inverter



IN	out.
V_{DD}	0V
0	

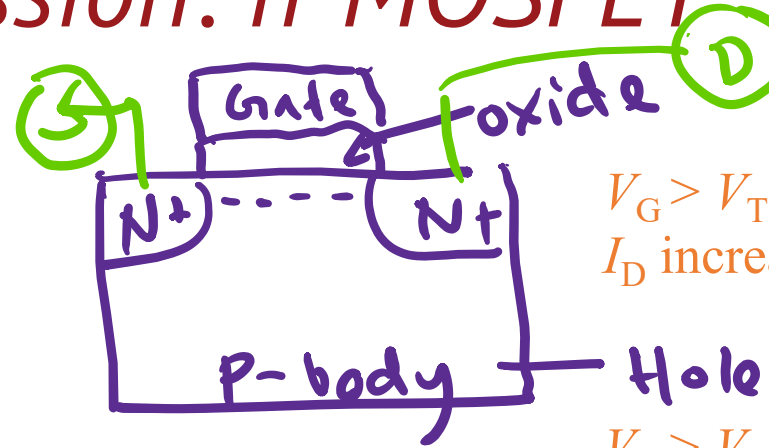
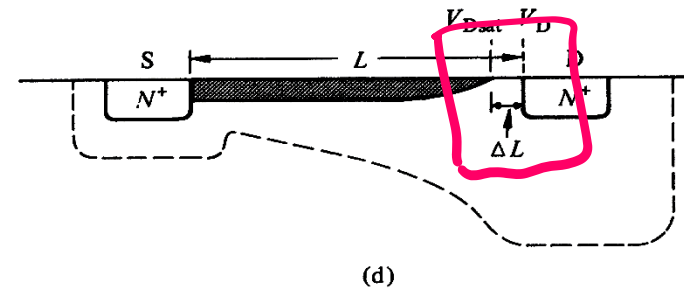
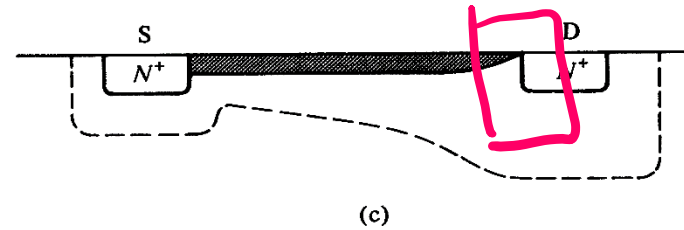
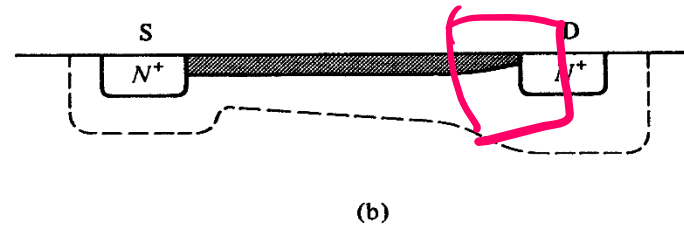
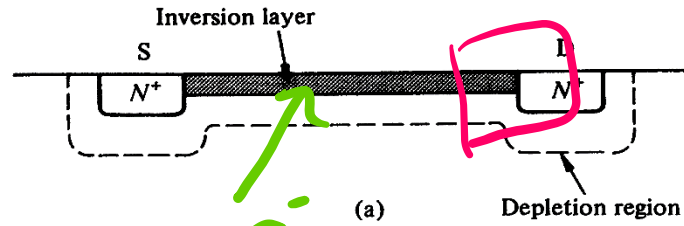
Hand-drawn truth table for a CMOS inverter. The input is labeled 'IN' and the output is labeled 'out.'. The first row shows the input V_{DD} (boxed) resulting in an output of 0V (green). The second row shows the input 0 resulting in an output of 0. A vertical arrow on the right indicates the output range from V_{DD} to GND.

CMOS (Complementary MOS) Inverter



IN	out.
V_{DD}	0V
0	V_{DD}

Qualitative discussion: n-MOSFET



$V_G > V_T$; $V_{DS} \approx 0$
 I_D increases with V_{DS}

$V_{DS} = 0$

Holes → majority, electrons - minority
 $V_G > V_T$; V_{DS} small, > 0
 I_D increases with V_{DS} , but rate of increase decreases.

V_{DS} small

$V_G > V_T$; $V_{DS} \approx \text{pinch-off}$
 I_D reaches a saturation value, $I_{D,sat}$
The V_{DS} value is called $V_{DS,sat}$

V_{DS} pinch off

$V_G > V_T$; $V_{DS} > V_{DS,sat}$
 I_D does not increase further, saturation region.

$V_{DS} > V_{DS,sat}$ pinch off volt

Quantitative I_D - V_{DS} Relationships – 1st attempt “Square Law”

$$V_{GS} - V_{DS} = V_T$$

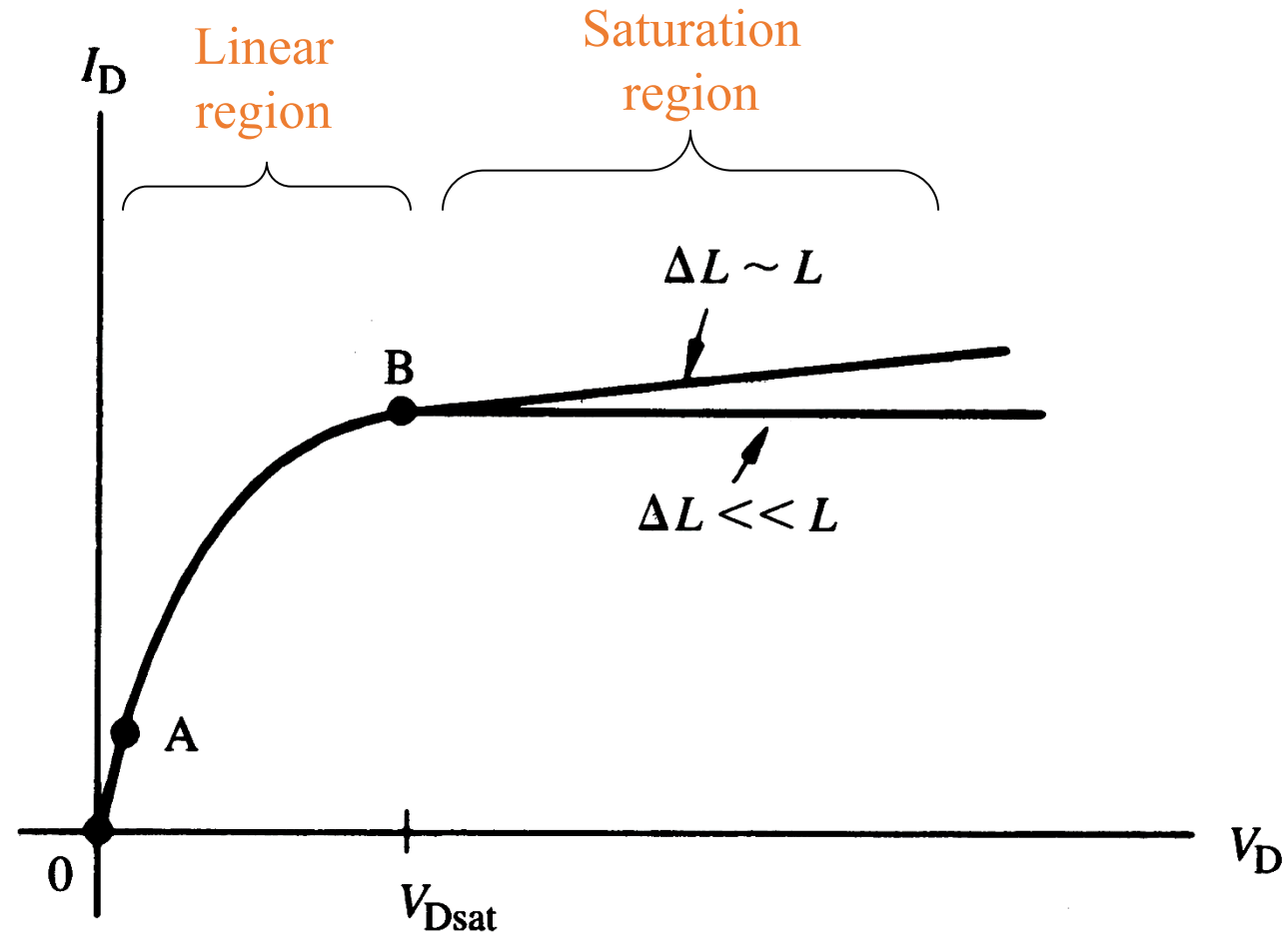
$$I_D = \mu C_{ox} \frac{W}{L} [(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2] \quad 0 < V_{DS} < V_{DS,sat} \quad ; \quad V_G > V_T$$

I_D will increase as V_{DS} is increased, but when $V_G - V_{DS} = V_T$, pinch-off occurs, and current saturates when V_{DS} is increased further. This value of V_{DS} is called $V_{DS,sat}$. i.e., $V_{DS,sat} = V_G - V_T$ and the current when $V_{DS} = V_{DS,sat}$ is called $I_{D,sat}$.

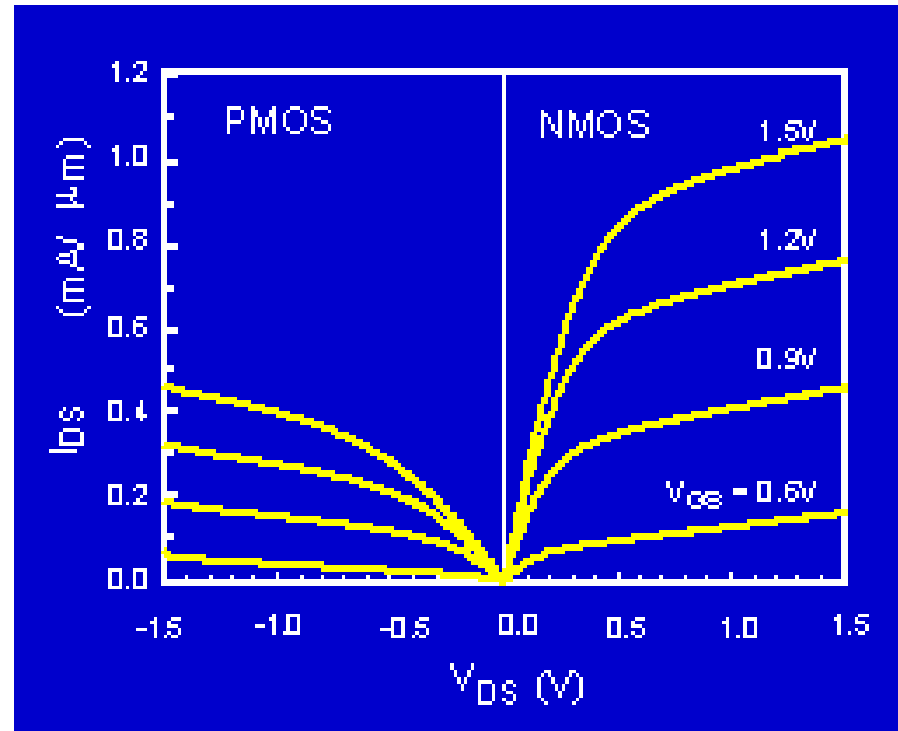
$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad V_D > V_{DS,sat} \quad ; \quad V_G > V_T$$

Here, C_{ox} is the oxide capacitance per unit area, $C_{ox} = \epsilon_{ox} / x_{ox}$

I_D - V_{DS} characteristics for n -MOSFET

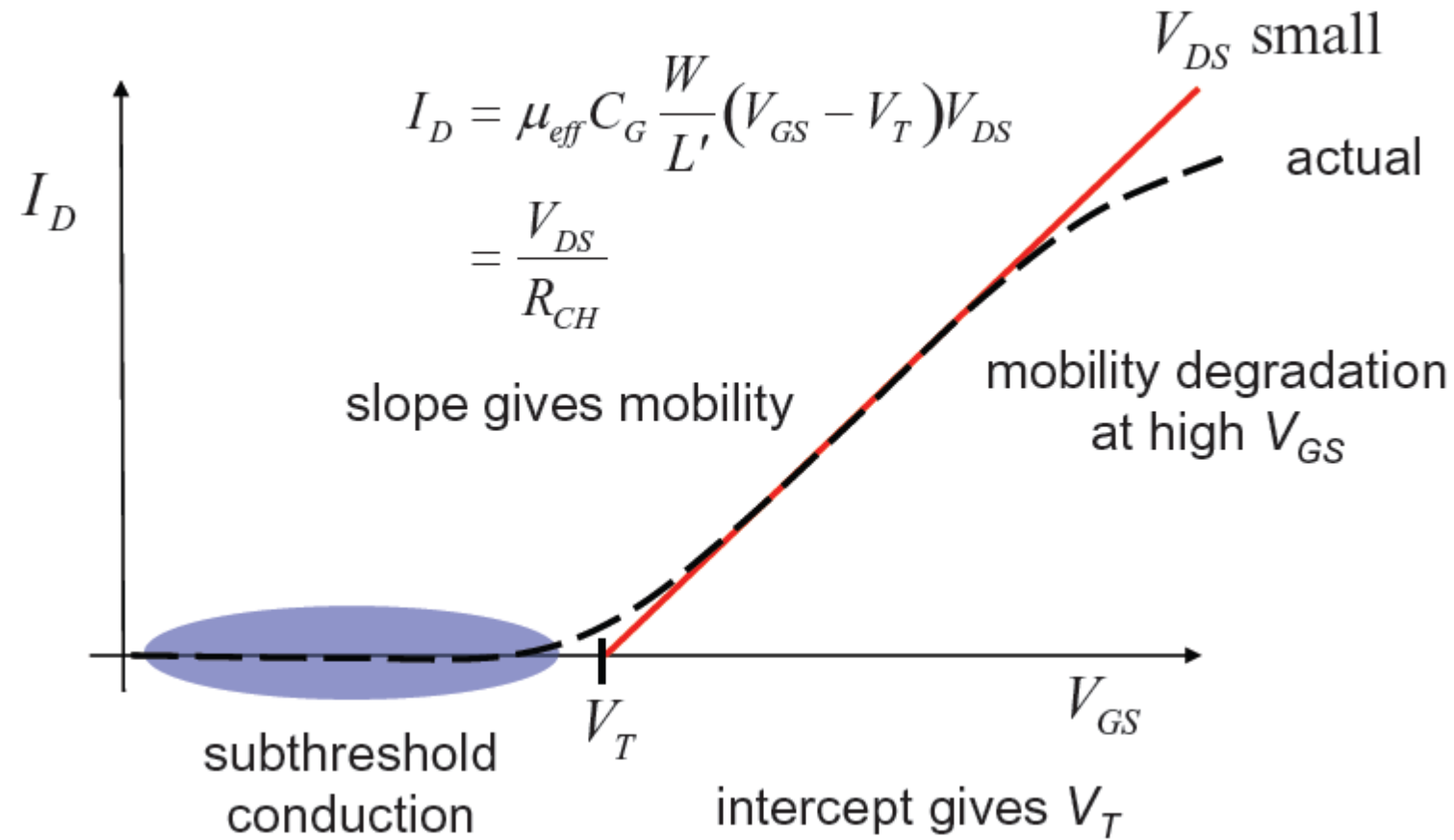


P-MOSFET N-MOSFET IV Characteristics

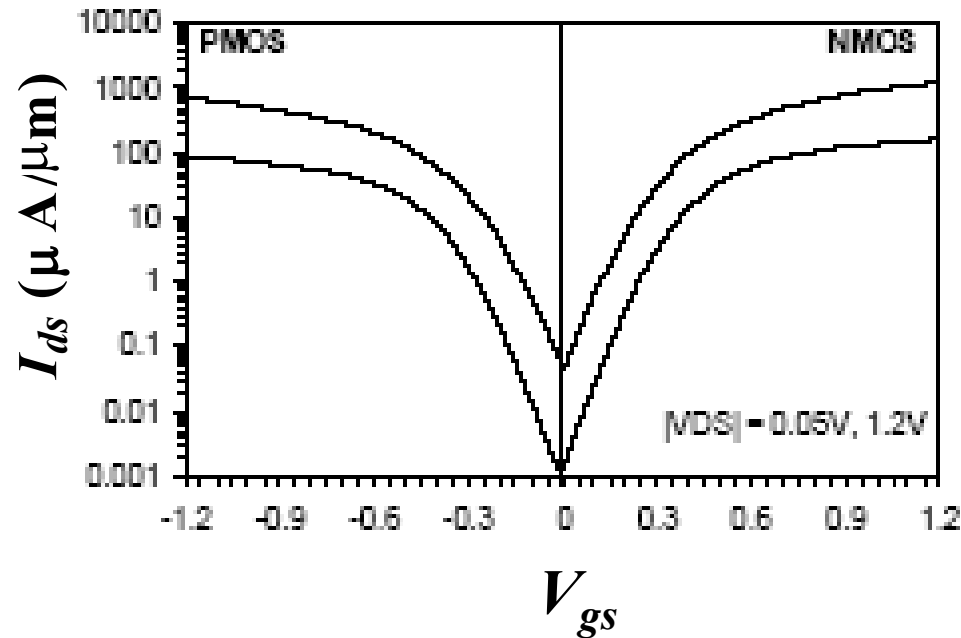


The PMOS IV is qualitatively similar to the NMOS IV, but the current is about half as large. Why?

Threshold and Subthreshold



- The leakage current that flows at $V_g < V_t$ is called the subthreshold current. Previously we had assumed that current is zero, but in reality that's not the case.



Intel, T. Ghani et al., IEDM 2003

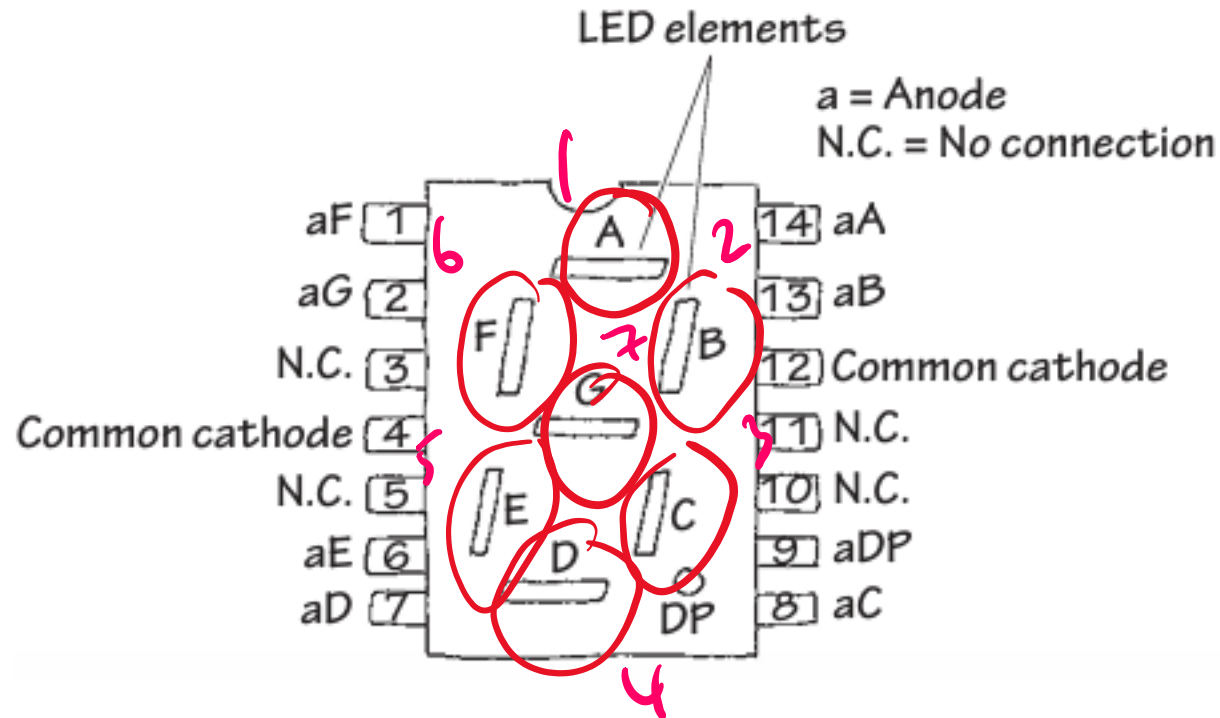
90nm technology.
Gate length: 45nm for
NMOS, 50nm for
PMOS

- The current at $V_{gs}=0$ and $V_{ds}=V_{dd}$ is called I_{off} .

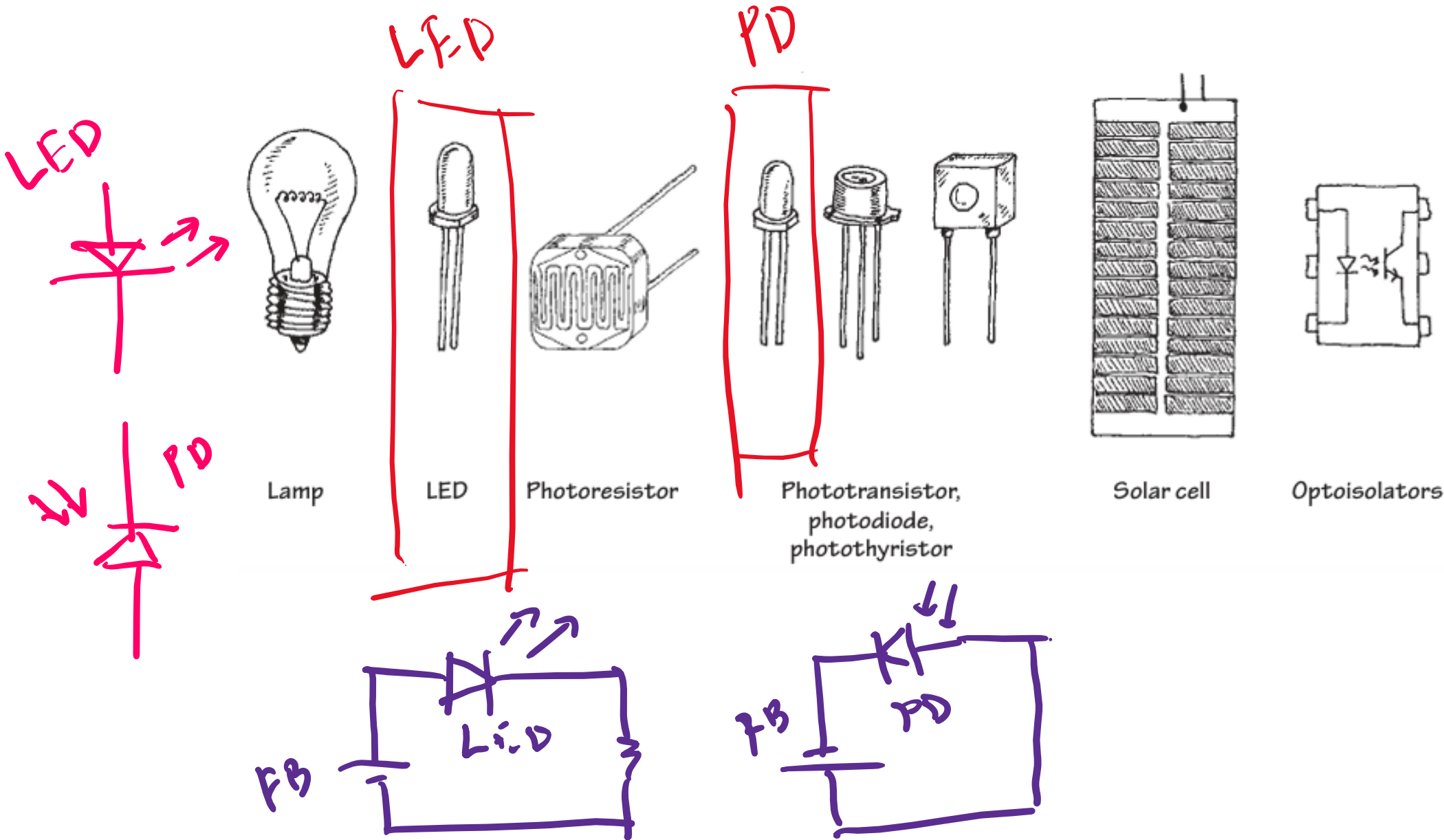
The seven-segment display

LED Displays

LED emit light
PD detect light

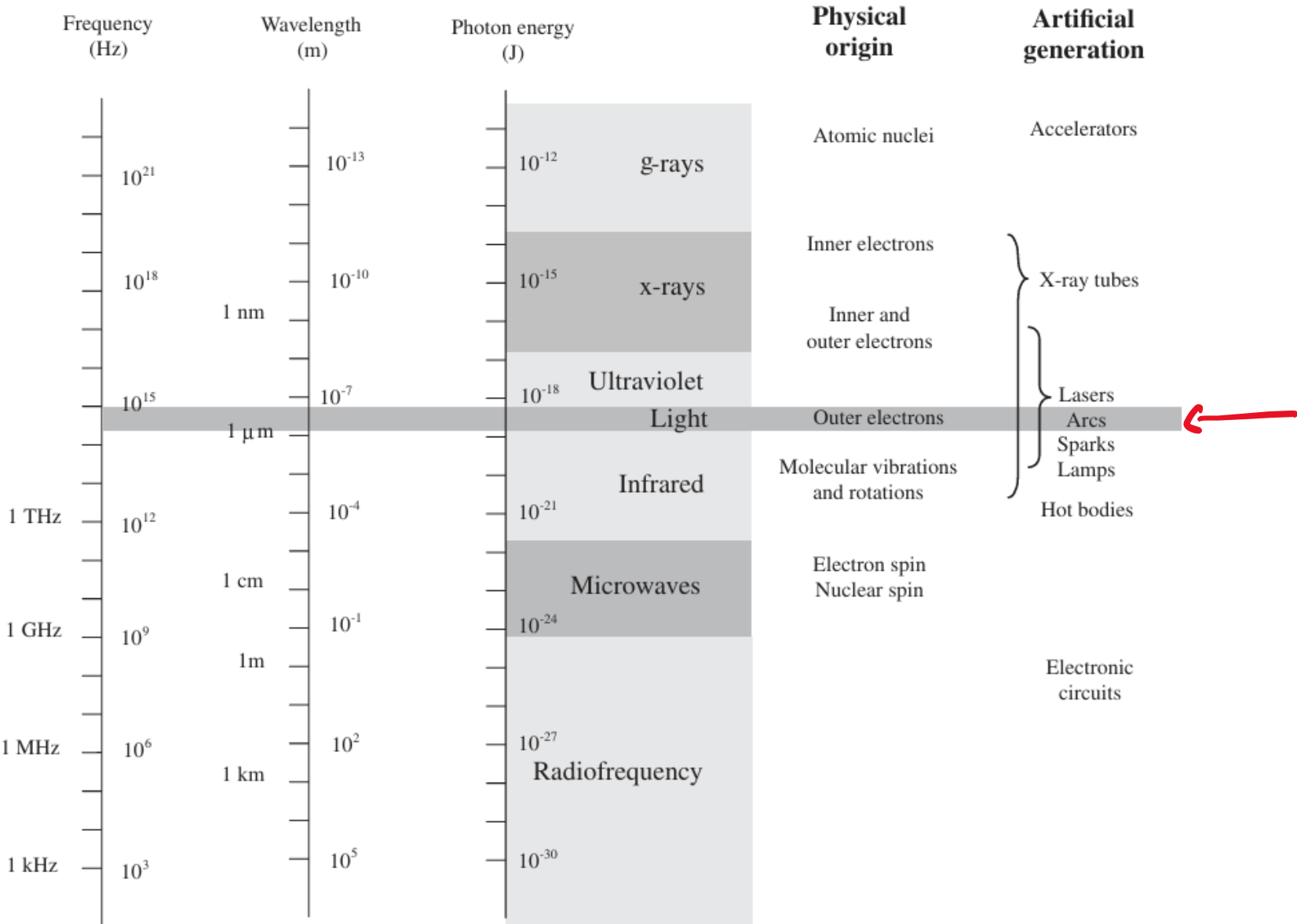


Light-based devices

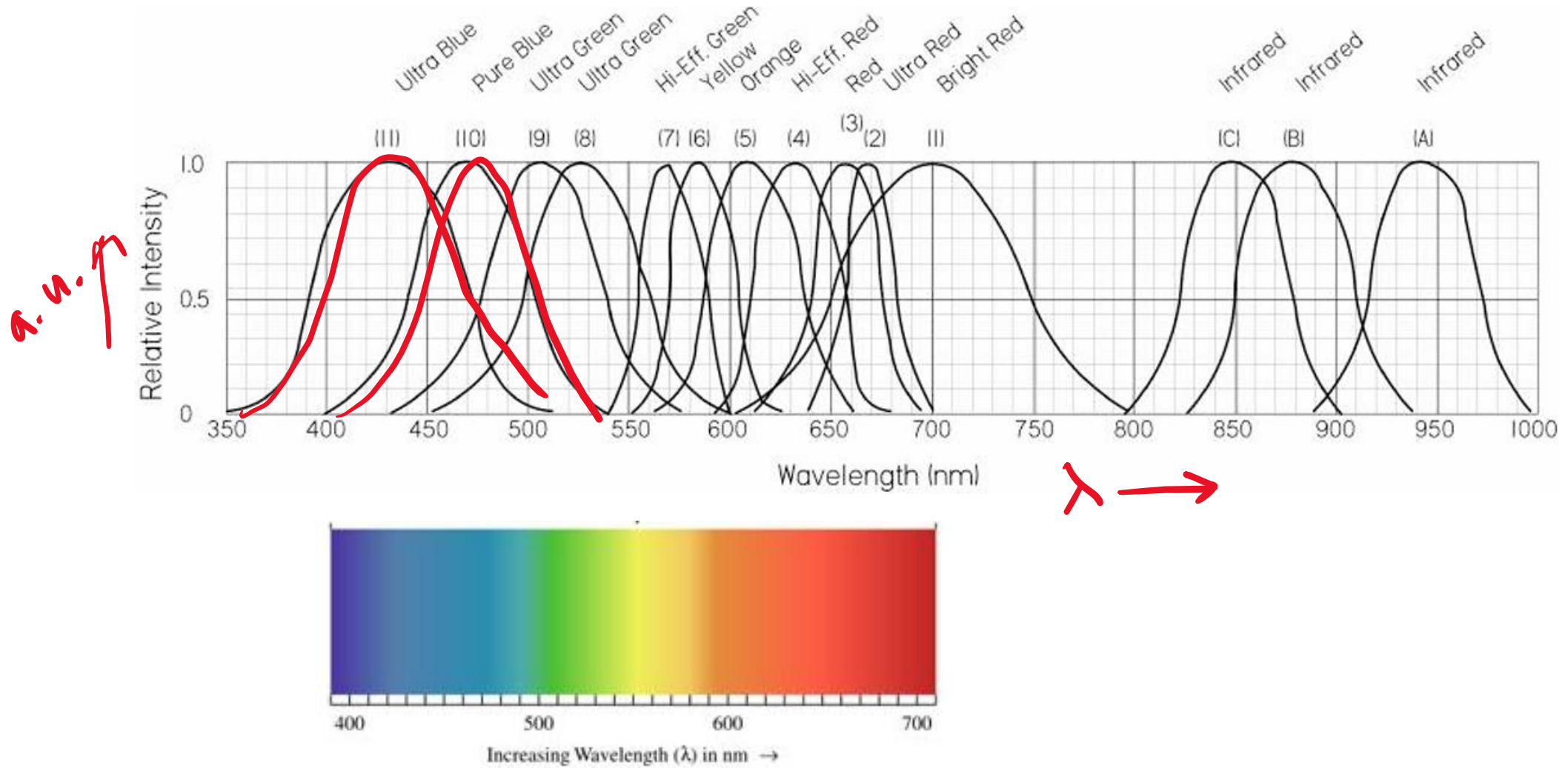


Artificially generating light

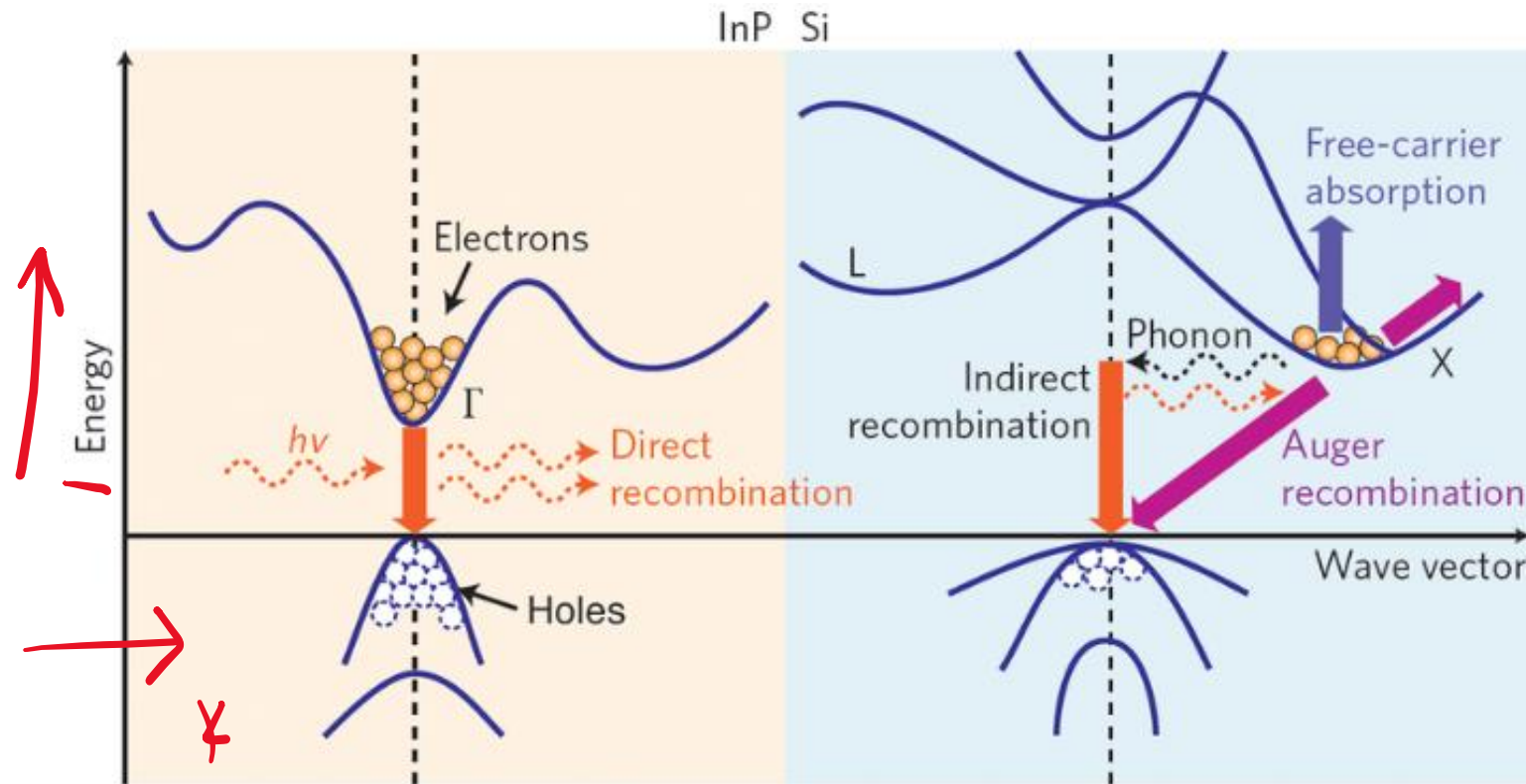
Electromagnetic Spectrum



LED emission spectra



Direct vs indirect semiconductors



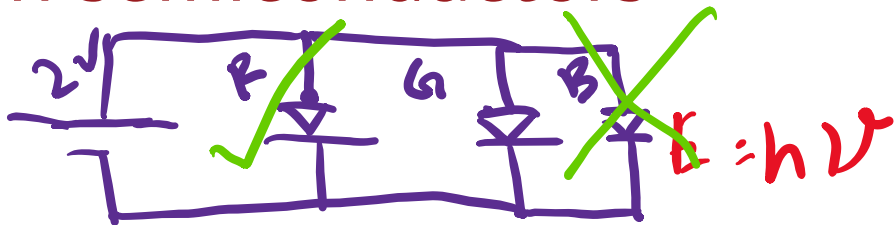
2 Energy band diagrams and major carrier transition processes in indium phosphide (direct bandgap) and silicon (indirect bandgap) crystals. Silicon, Germanium and some **III/V compounds like GaP and AlAs are indirect bandgap semiconductors**. InP, GaAs, GaN and other ternary (AlGaAs, InGaAs) and quaternary (InGaAsP, InAlAsP) compounds are direct bandgap semiconductors commonly used for photonic devices fabrication.

LEDs are made out of non-silicon semiconductors

700nm

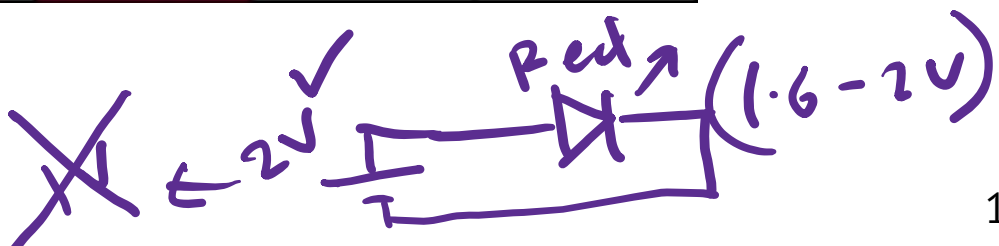
	Color	Wavelength [nm]	Semiconductor material
	Infrared	$\lambda > 760$	Gallium arsenide (GaAs) Aluminium gallium arsenide (AlGaAs)
	Red	$610 < \lambda < 760$	Aluminium gallium arsenide (AlGaAs) ← Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
	Orange	$590 < \lambda < 610$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
	Yellow	$570 < \lambda < 590$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
	Green	$500 < \lambda < 570$	Traditional green: Gallium(III) phosphide (GaP) ← Aluminium gallium indium phosphide (AlGaInP) Aluminium gallium phosphide (AlGaP) Pure green: Indium gallium nitride (InGaN) / Gallium(III) nitride (GaN)
	Blue	$450 < \lambda < 500$	Zinc selenide (ZnSe) Indium gallium nitride (InGaN) Silicon carbide (SiC) as substrate Silicon (Si) as substrate—under development
	Violet	$400 < \lambda < 450$	Indium gallium nitride (InGaN)
	Purple	multiple types	Dual blue/red LEDs, blue with red phosphor, or white with purple plastic
	Ultraviolet	$\lambda < 400$	Diamond (235 nm) Boron nitride (215 nm) Aluminium nitride (AlN) (210 nm) Aluminium gallium nitride (AlGaIn) Aluminium gallium indium nitride (AlGaInN)—down to 210 nm
	Pink	multiple types	Blue with one or two phosphor layers: yellow with red, orange or pink phosphor added afterwards, or white with pink pigment or dye.
	White	Broad spectrum	Blue/UV diode with yellow phosphor

UV
400

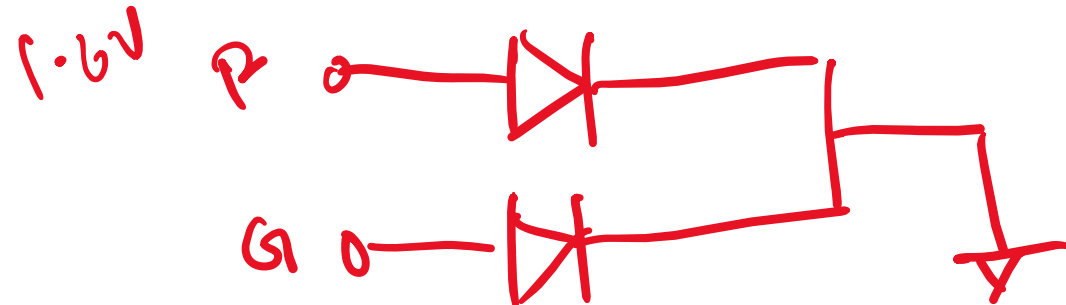
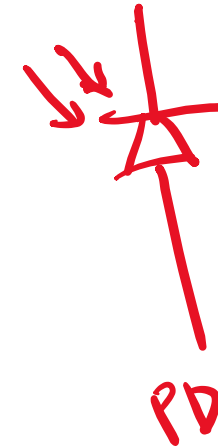
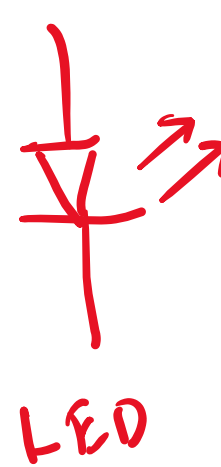
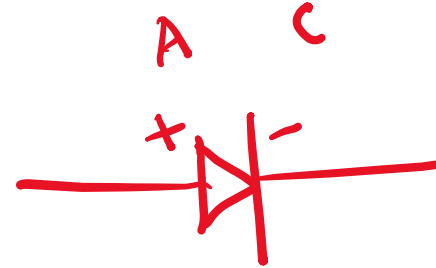
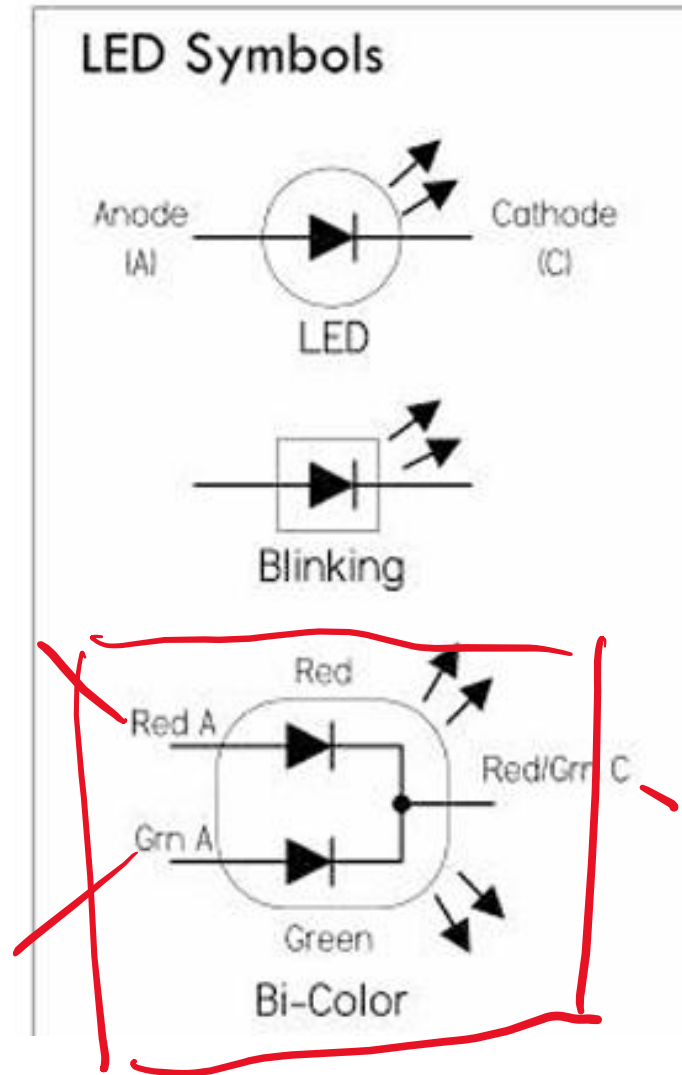


Color LED	Name Color	Wavelength nm = 1x10 ⁻⁹	Voltage Drop (Forward Voltage)
	White	395 - 530 nm	3 - 5 V
	Ultraviolet	< 400 nm	3.1 - 4.4 V
	Violet	400 - 450 nm	2.8 - 4.0 V
	Blue	450 - 500 nm	2.5 - 3.7 V
	Green	500 - 570 nm	1.9 - 4.0 V
	Yellow	570 - 590 nm	2.1 - 2.2 V
	Orange	590 - 610 nm	2.0 - 2.1 V
	Red	610 - 760 nm	1.6 - 2.0 V
	Infrared	> 760 nm	< 1.9 V

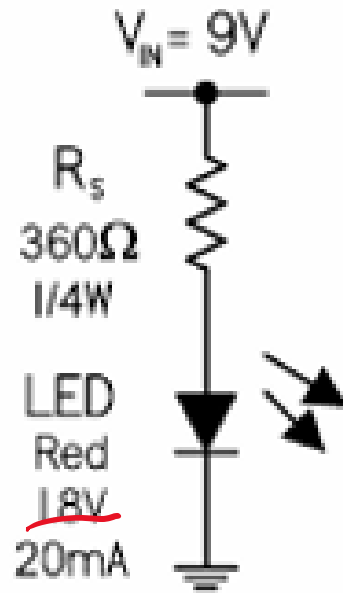
Blue
2.5-3.7



LED symbols



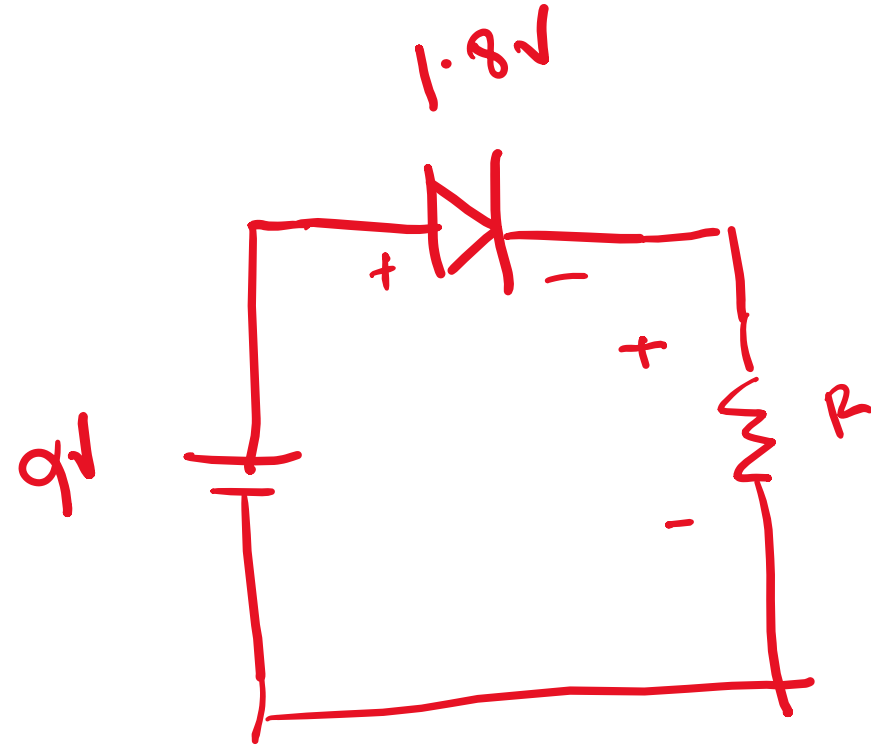
a. LED Current Limiting



$$R_S = \frac{V_{IN} - V_{LED}}{I_{LED}}$$

V_{LED} = LED forward voltage

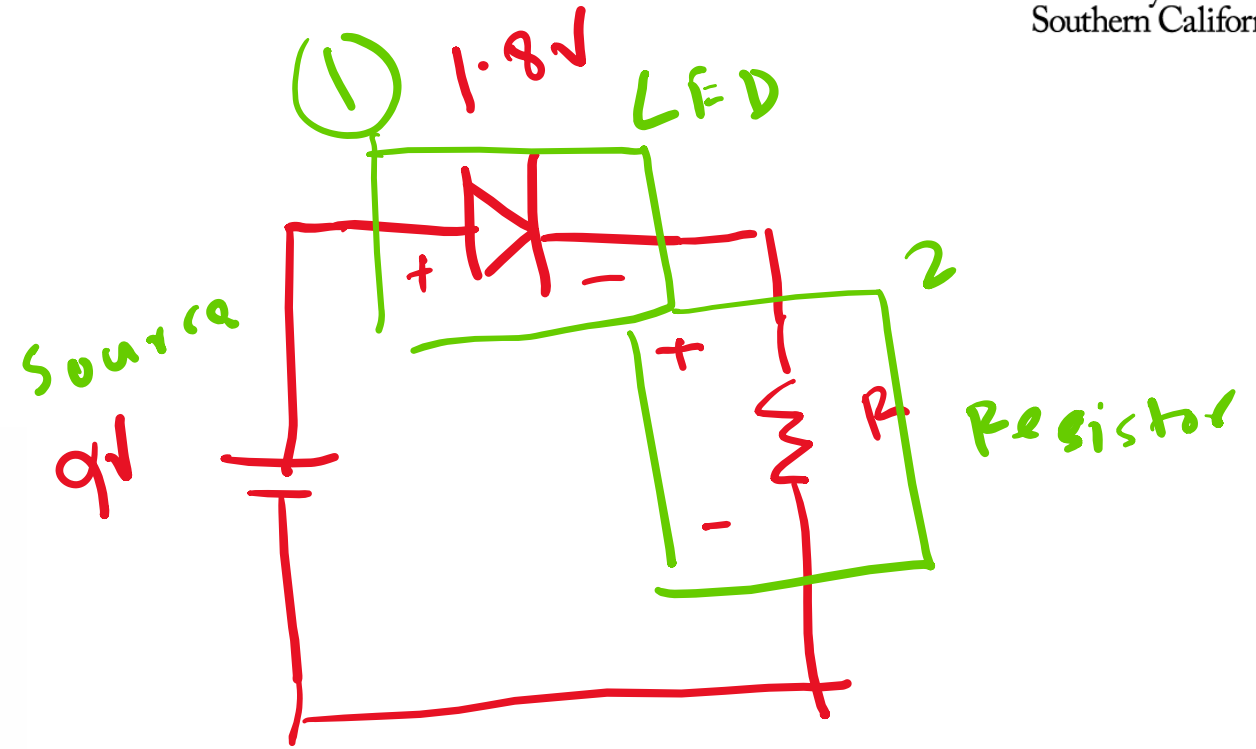
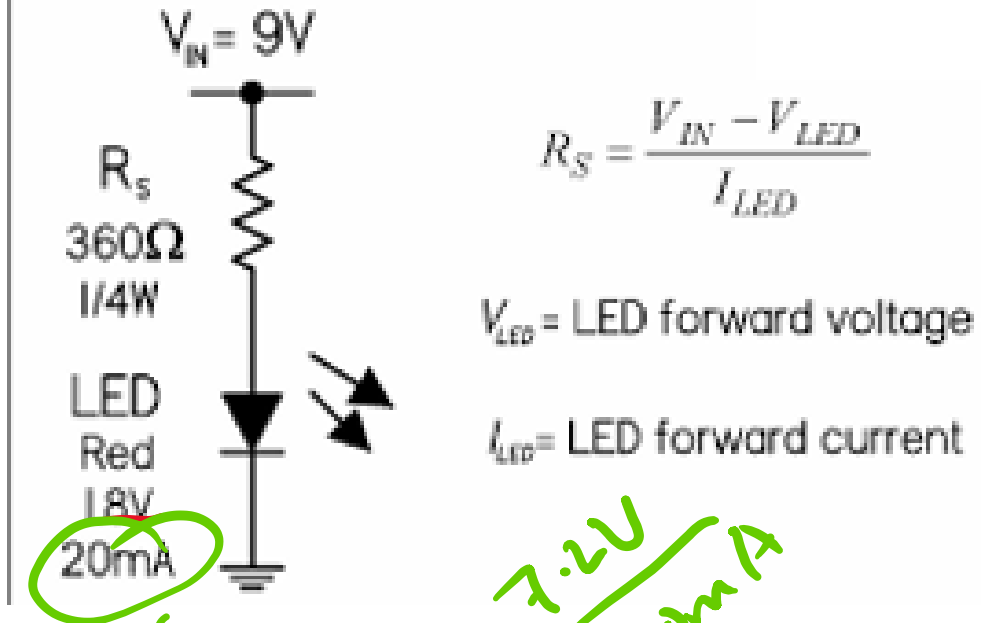
I_{LED} = LED forward current



$$V_S = V_{LED} + I R$$
$$9V = 1.8 + 20mA \cdot R$$
$$7.2V = 20mA \cdot R$$
$$R = \frac{7.2}{20m} = 360\Omega$$

LED driver circuit

a. LED Current Limiting



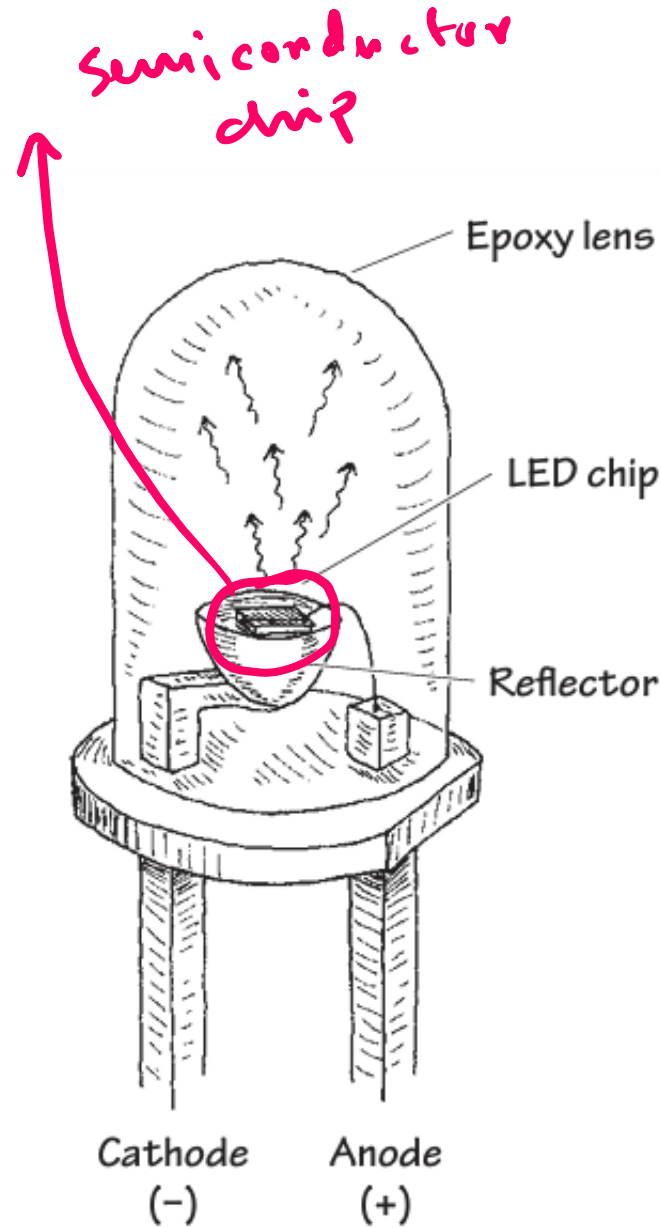
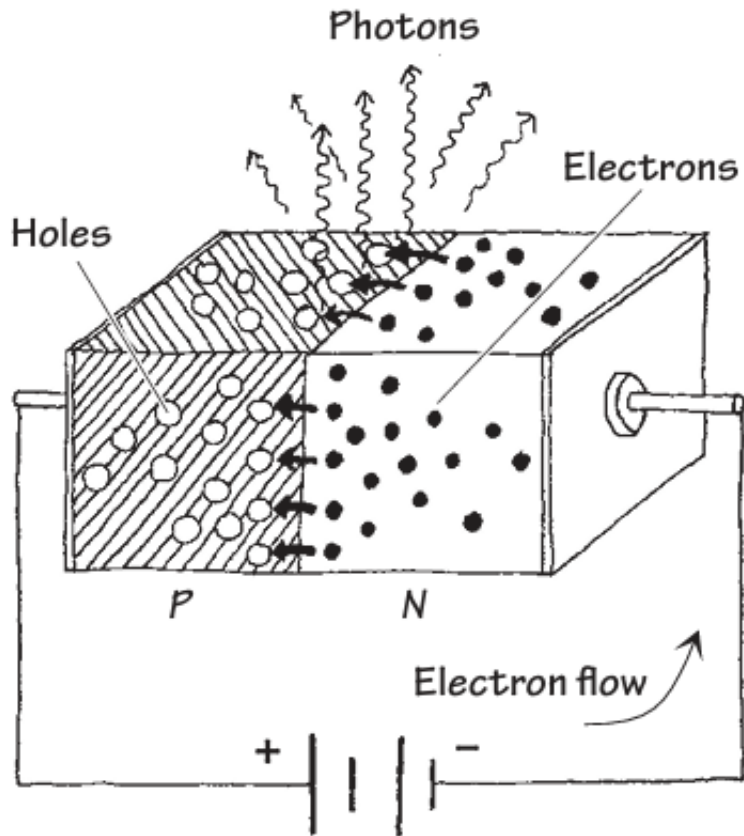
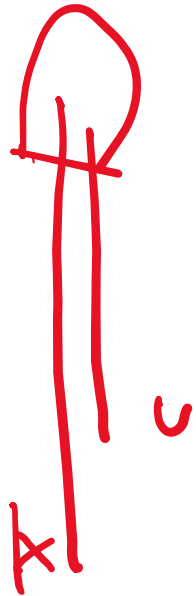
$$V_S = V_{drop,1} + V_{drop,2}$$

$$9V = 1.8V + IR$$

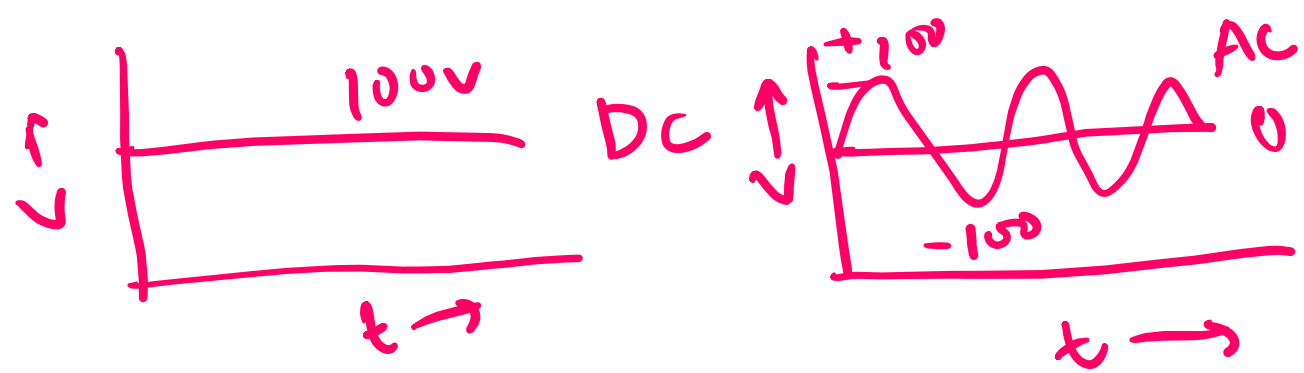
$$7.2V = I \times R$$

$$\rightarrow 20mA \times R$$

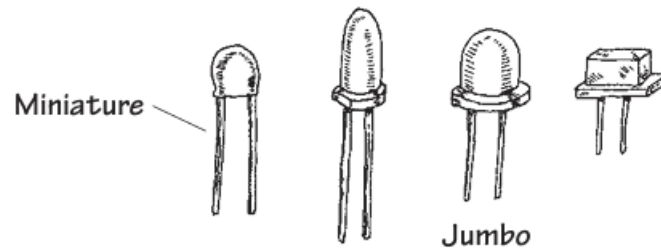
LED is a diode



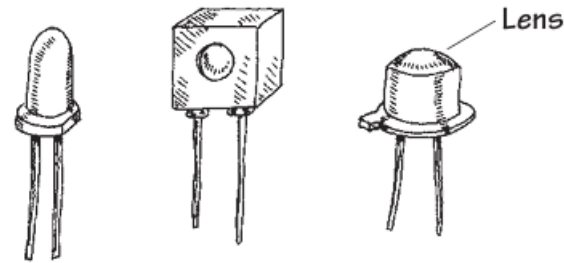
LED packages



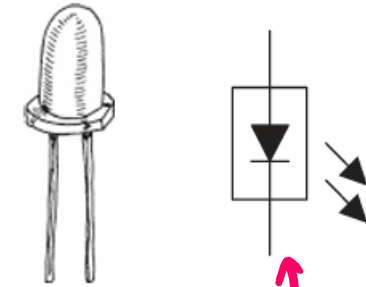
Visible-Light LEDs



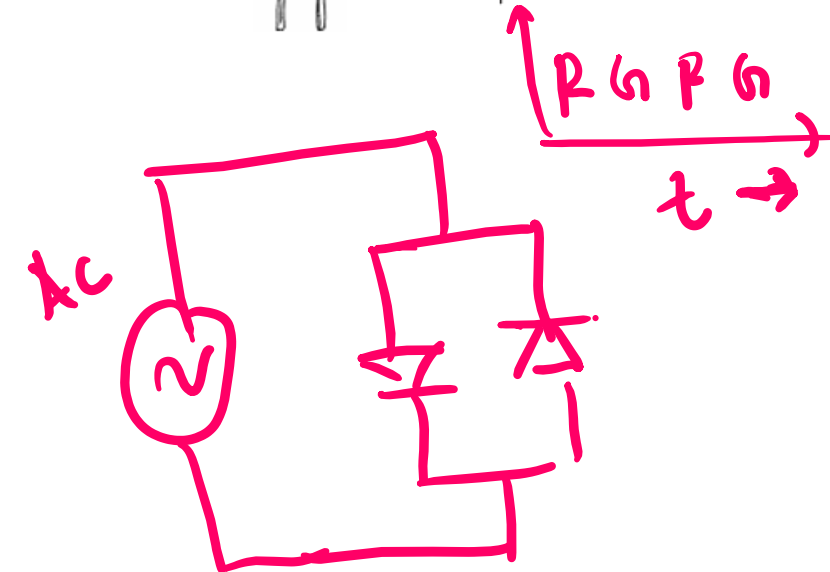
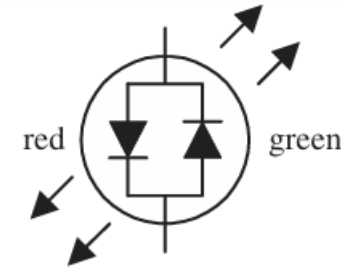
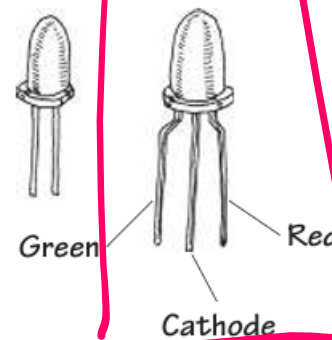
Infrared LEDs



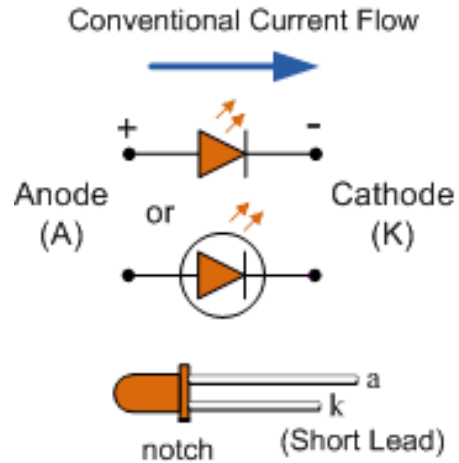
Blinking LEDs



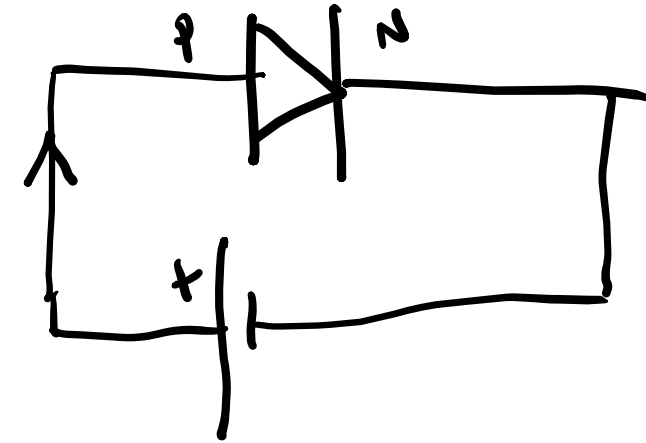
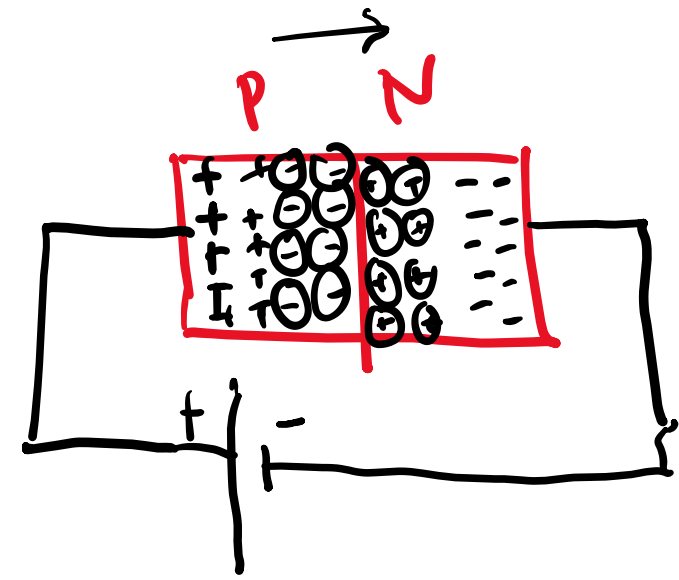
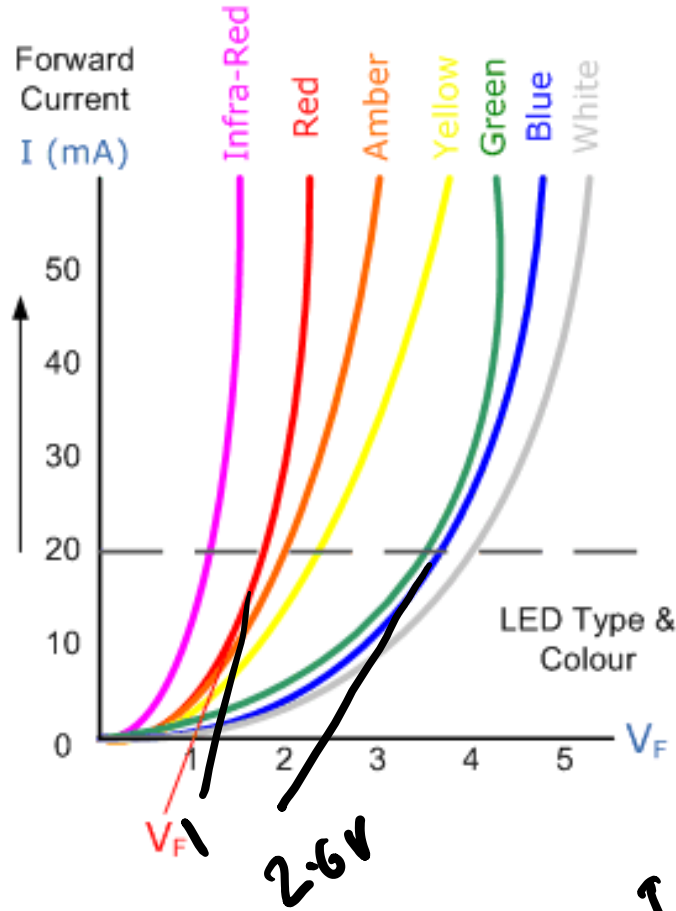
Multicolor LEDs



IV characteristics for LEDs



LED and its
I-V Characteristics



I_D

V_D

forward bias

Blue LED won the 2014 Nobel in Physics

© The Nobel Foundation. Photo: Lovisa Engblom.



The Royal Swedish Academy of Sciences has decided to award the

2014 NOBEL PRIZE IN PHYSICS

to:

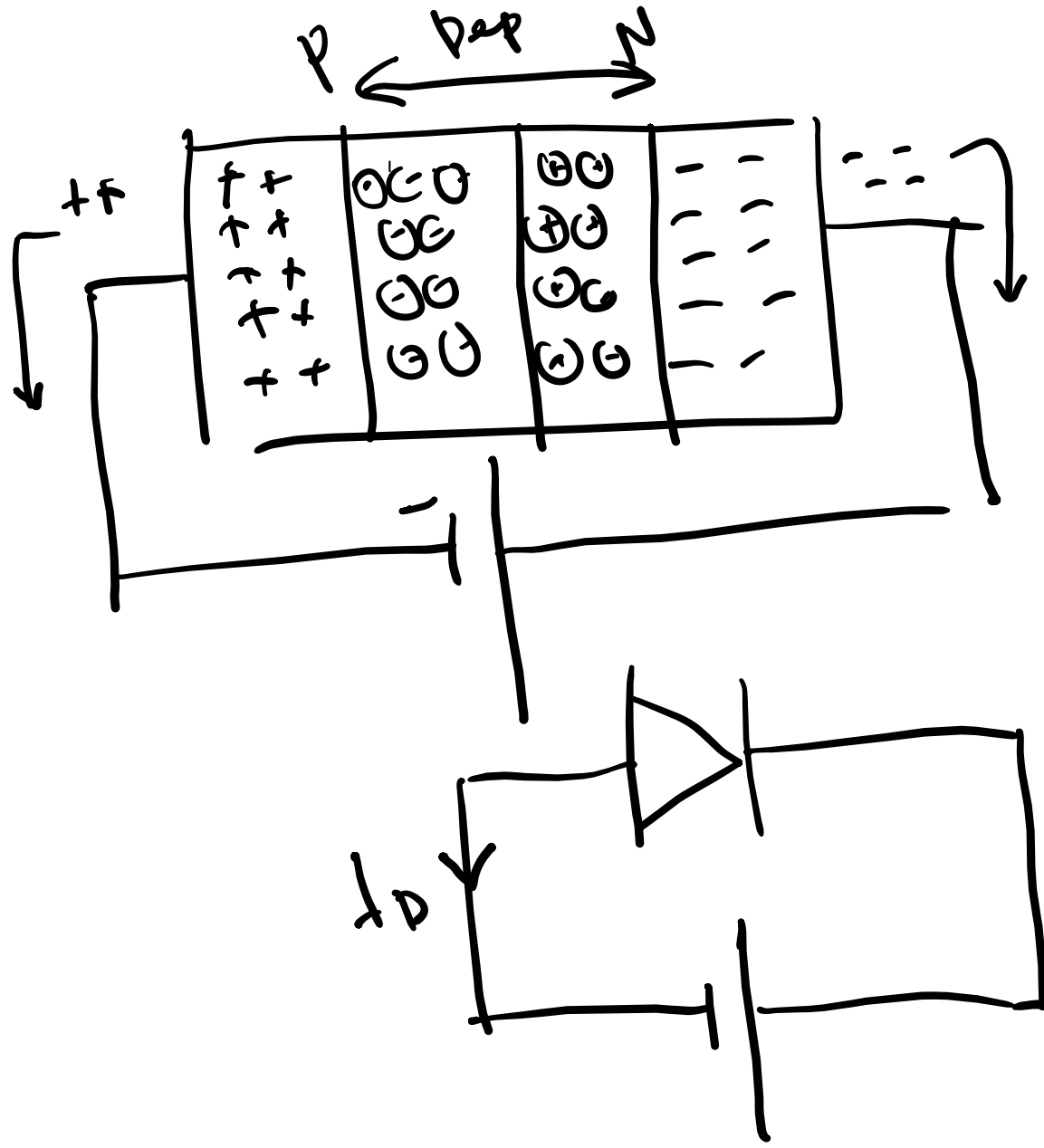


**Isamu Akasaki, Hiroshi Amano
and Shuji Nakamura**

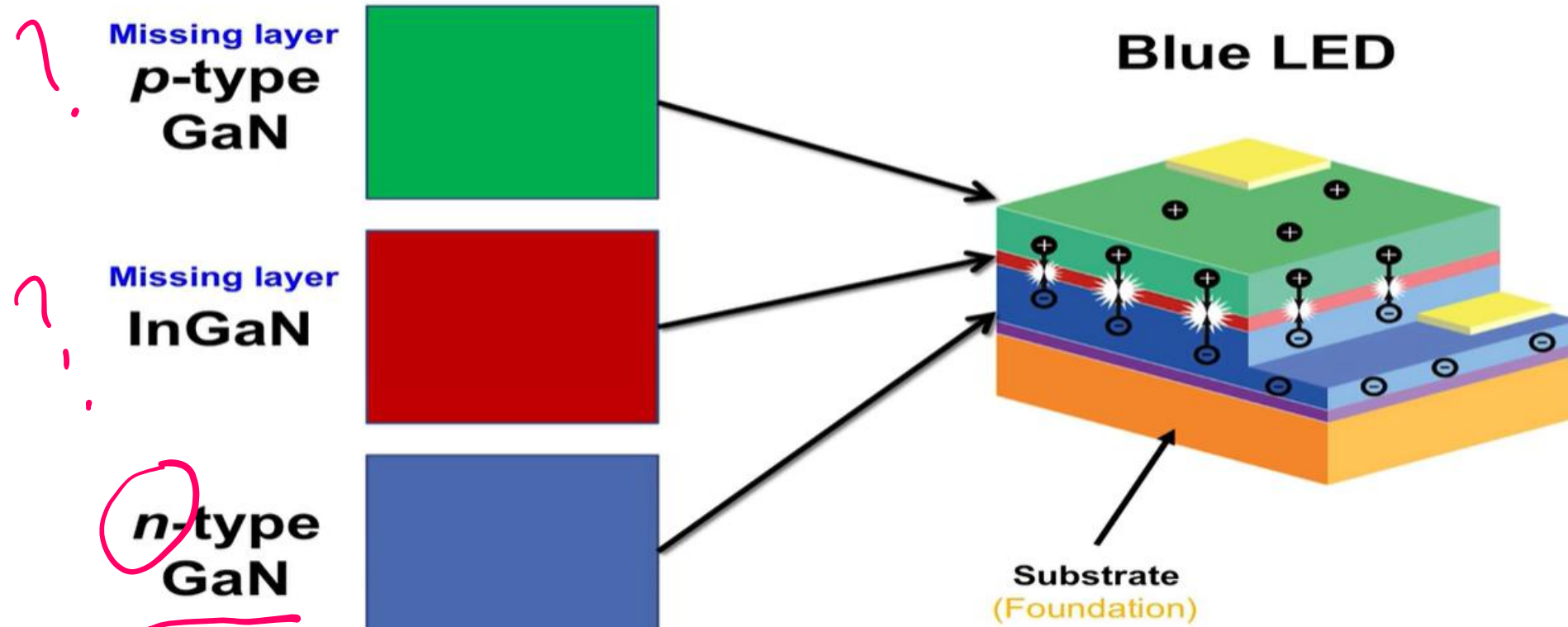
"for the invention of efficient blue light-emitting diodes which has enabled bright and energy-saving white light sources"

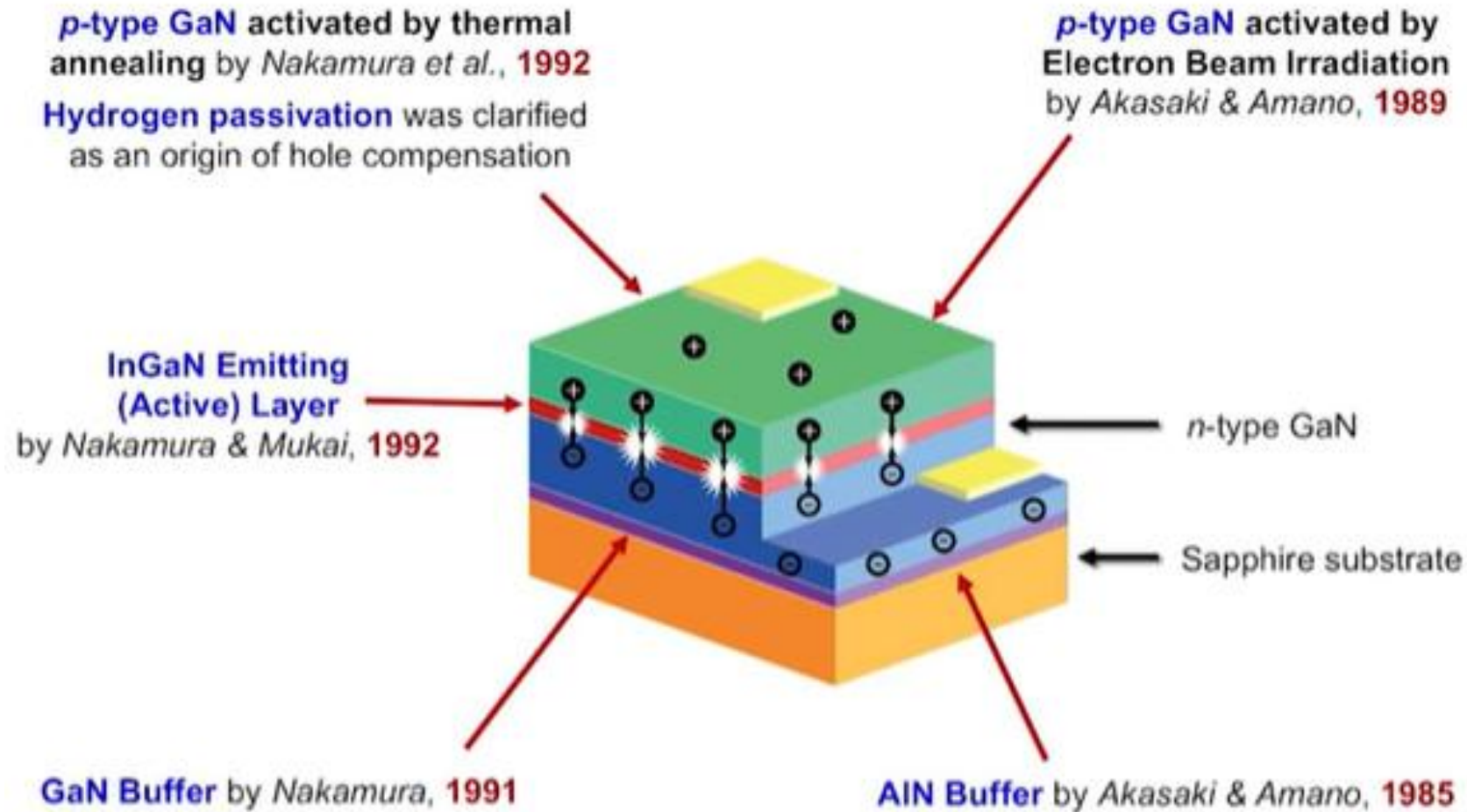
 **Nobelprize.org**
The Official Web Site of the Nobel Prize

Illustrations: Niklas Elmehed © Nobel Media AB

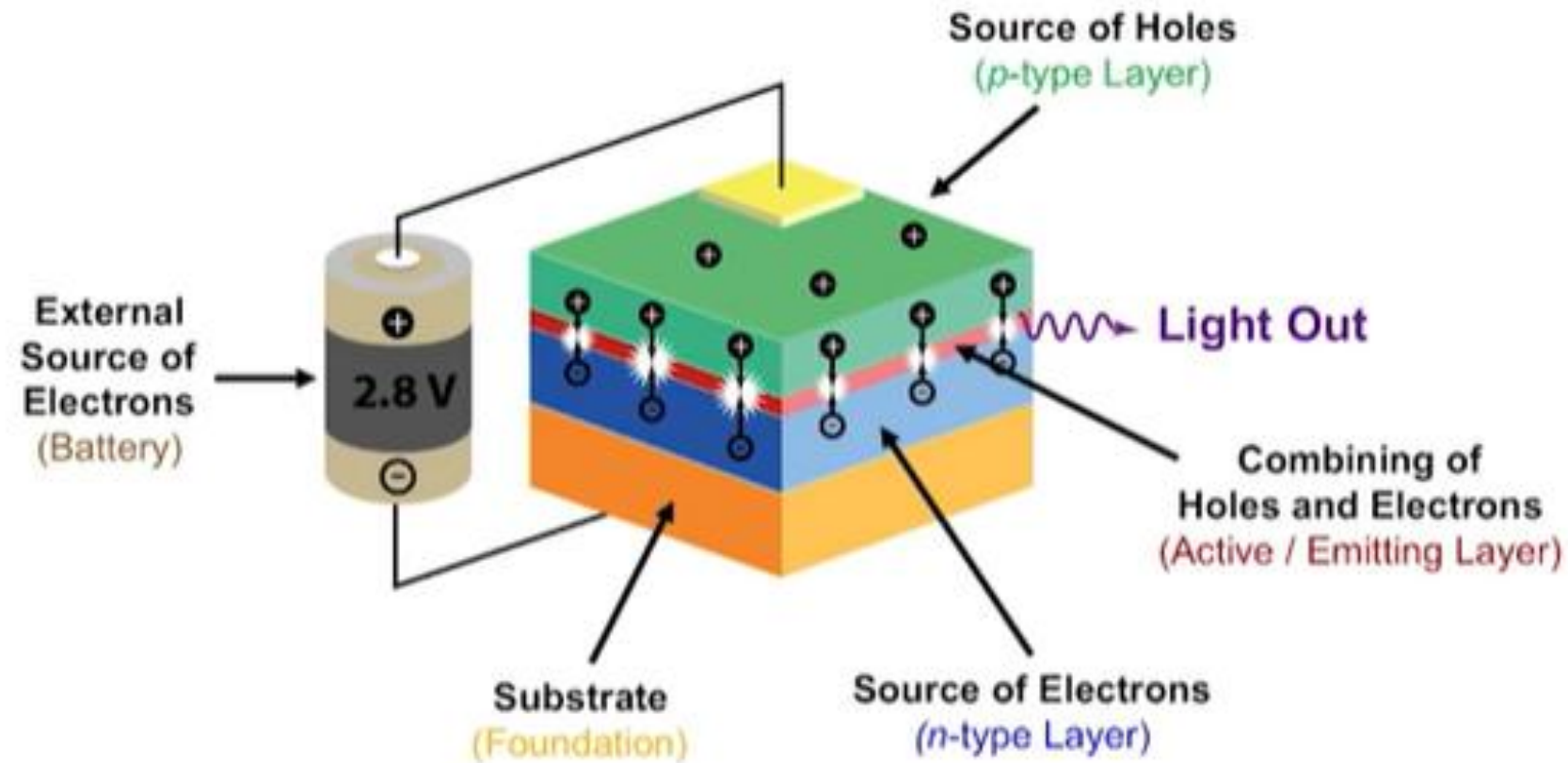


Steps to get the Blue LED

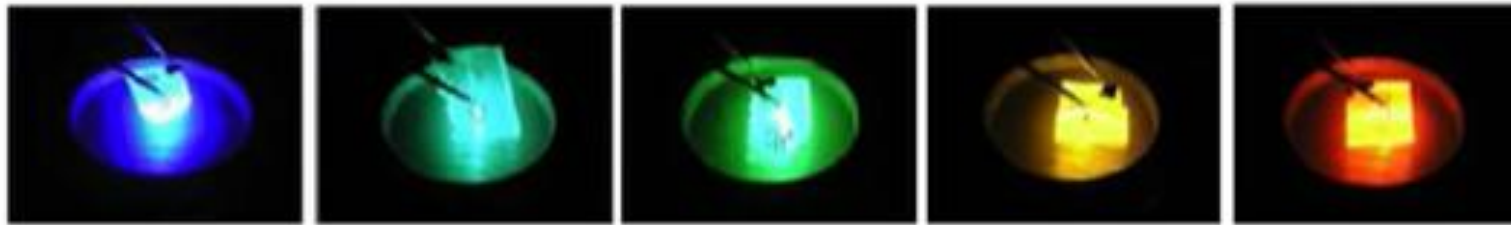
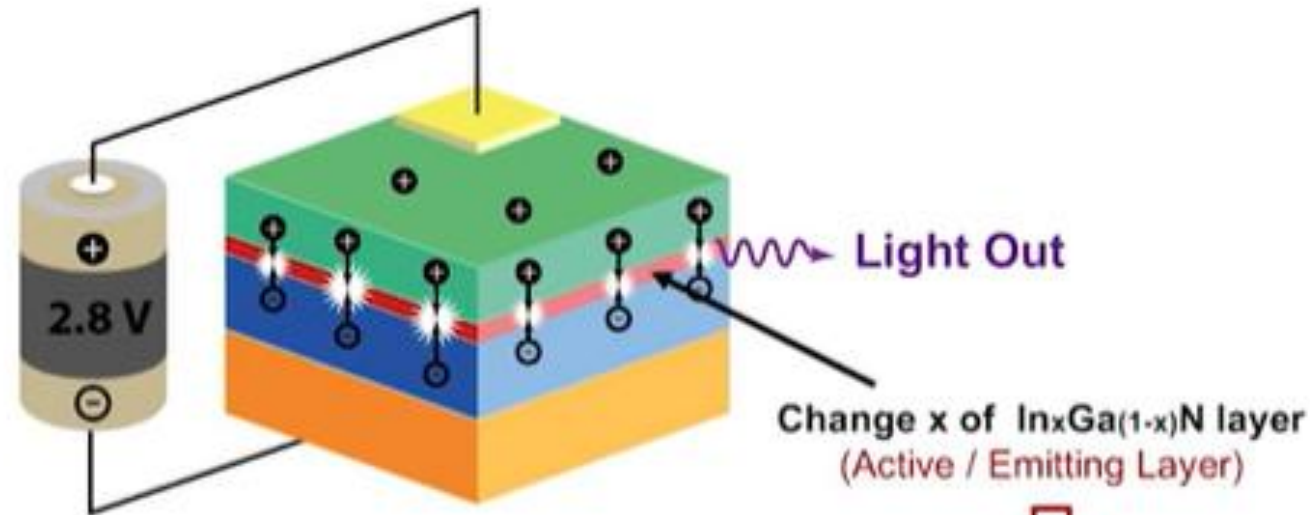




LED colors – single color from on LED



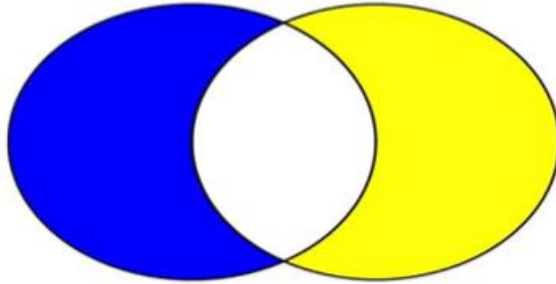
Changing emission colors – can you change color?



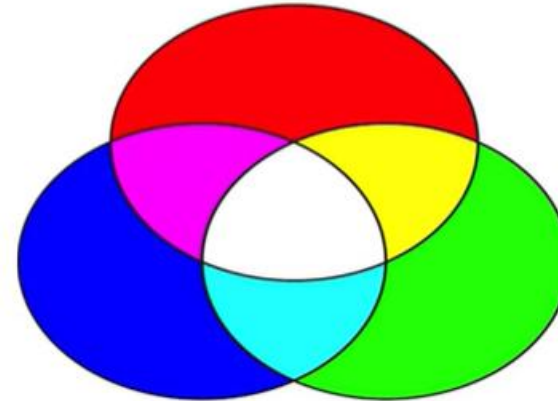
How to get white light

One LED can only produce **one color**
(red, orange, yellow, green, blue, or violet)

To achieve white light, need to **combine colors**:

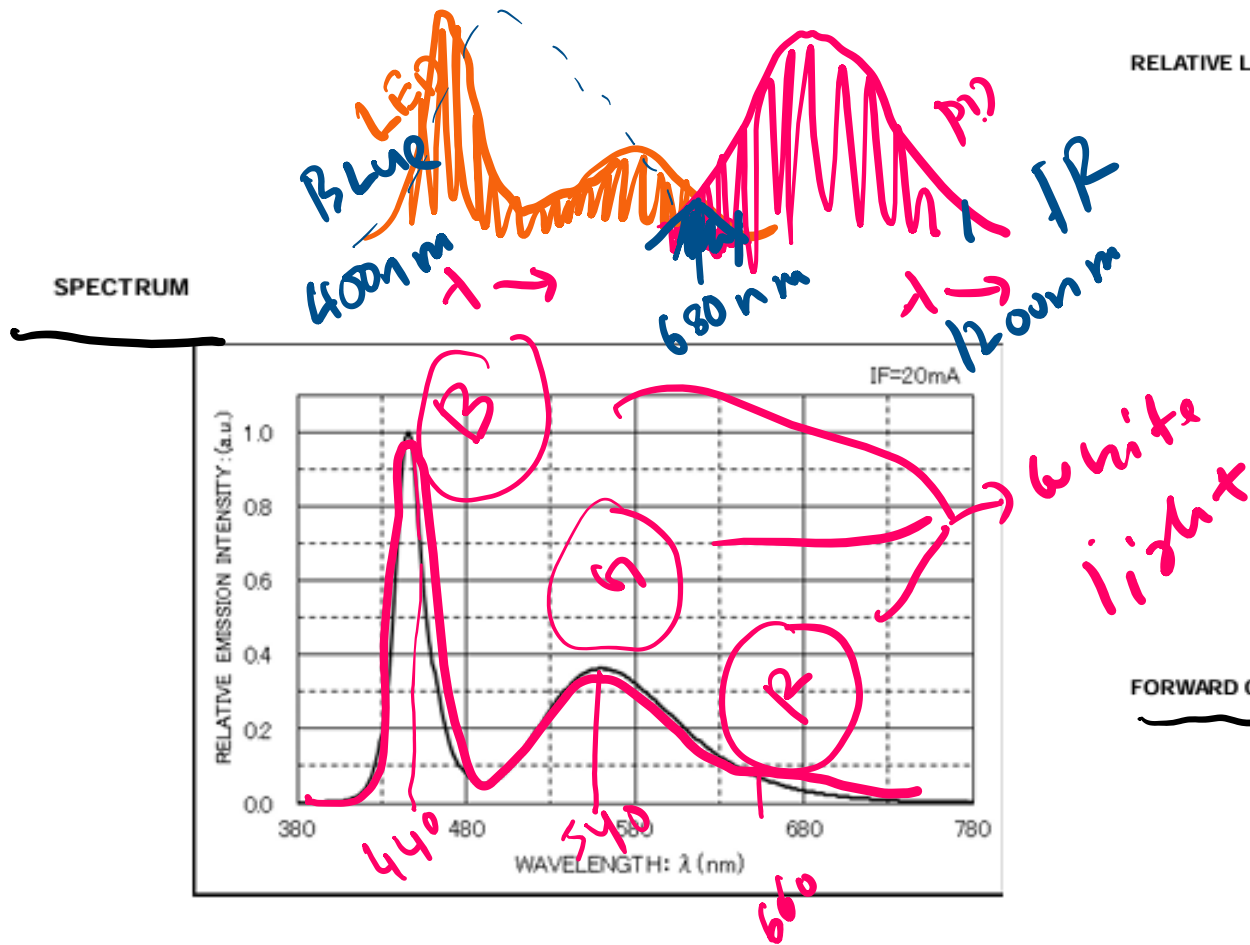


Blue + Yellow
(Easiest)

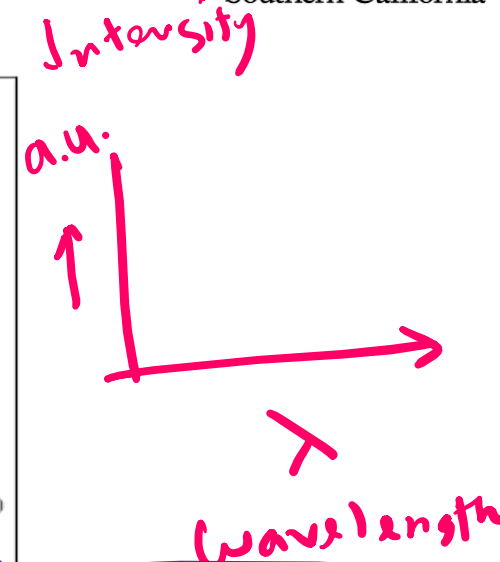
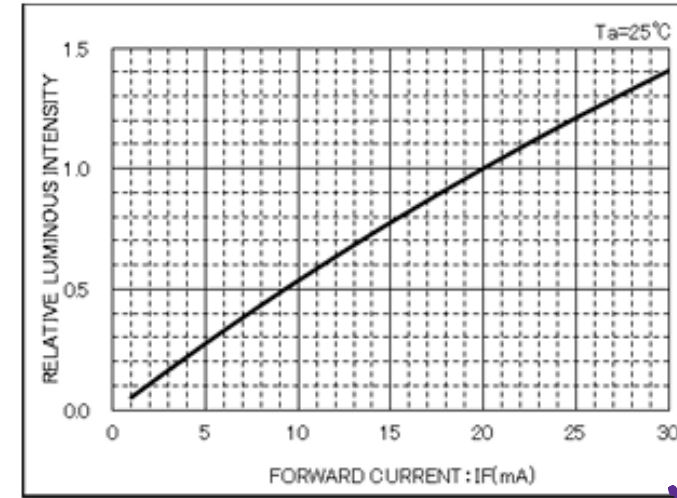


Blue + Green + Red
(Highest Quality)

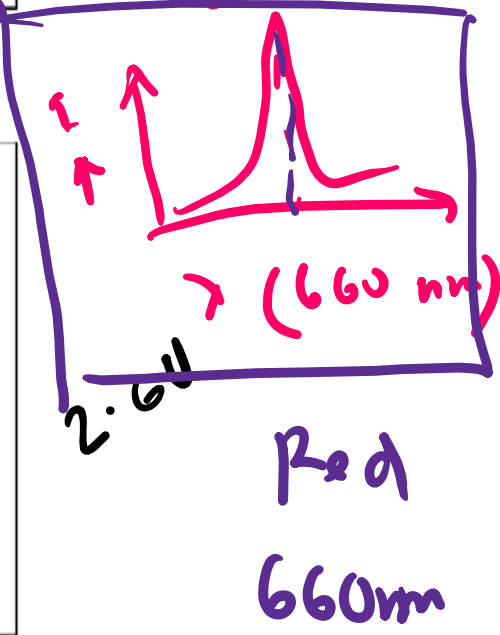
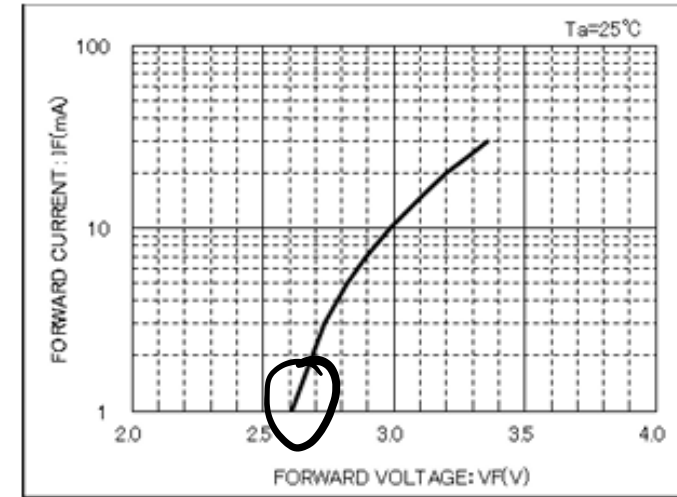
The LED in our demo board



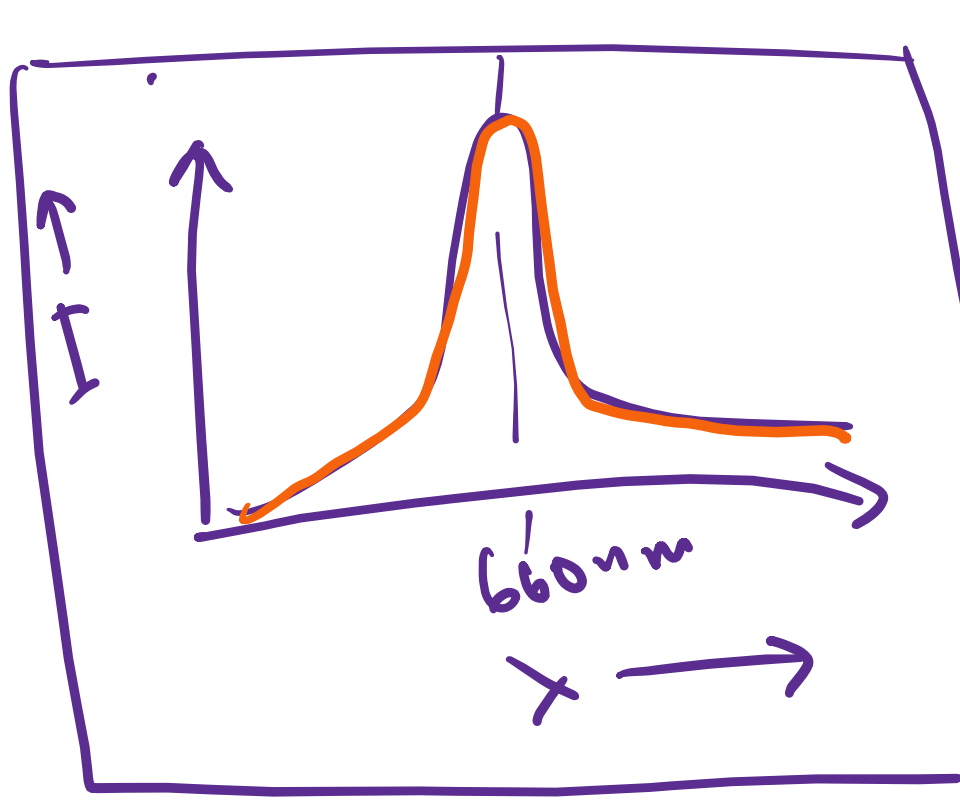
RELATIVE LUMINOUS INTENSITY - FORWARD CURRENT



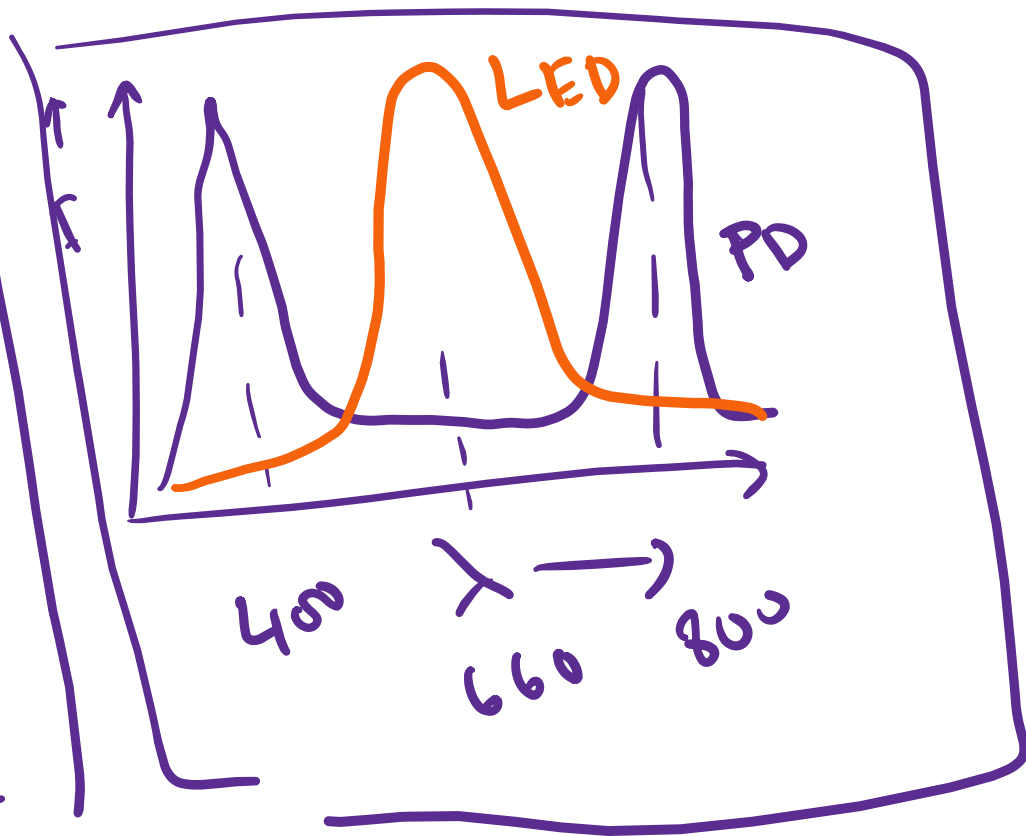
FORWARD CURRENT - FORWARD VOLTAGE



https://fscdn.rohm.com/en/products/databook/datasheet/opt/o/led/chip_mono/csl0416wbcw1-e.pdf



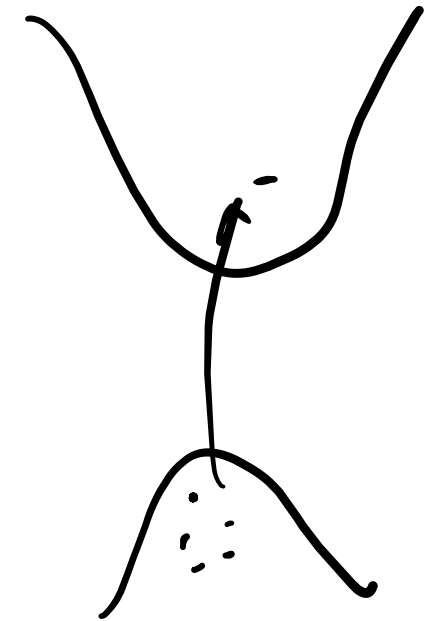
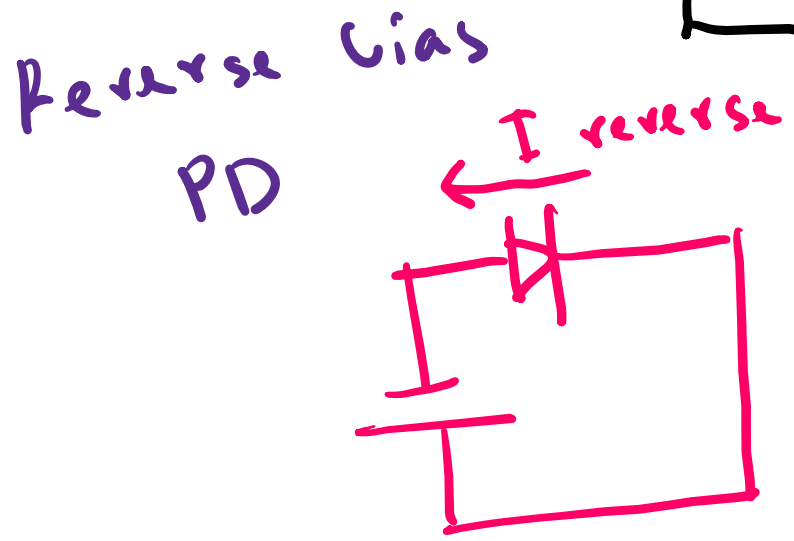
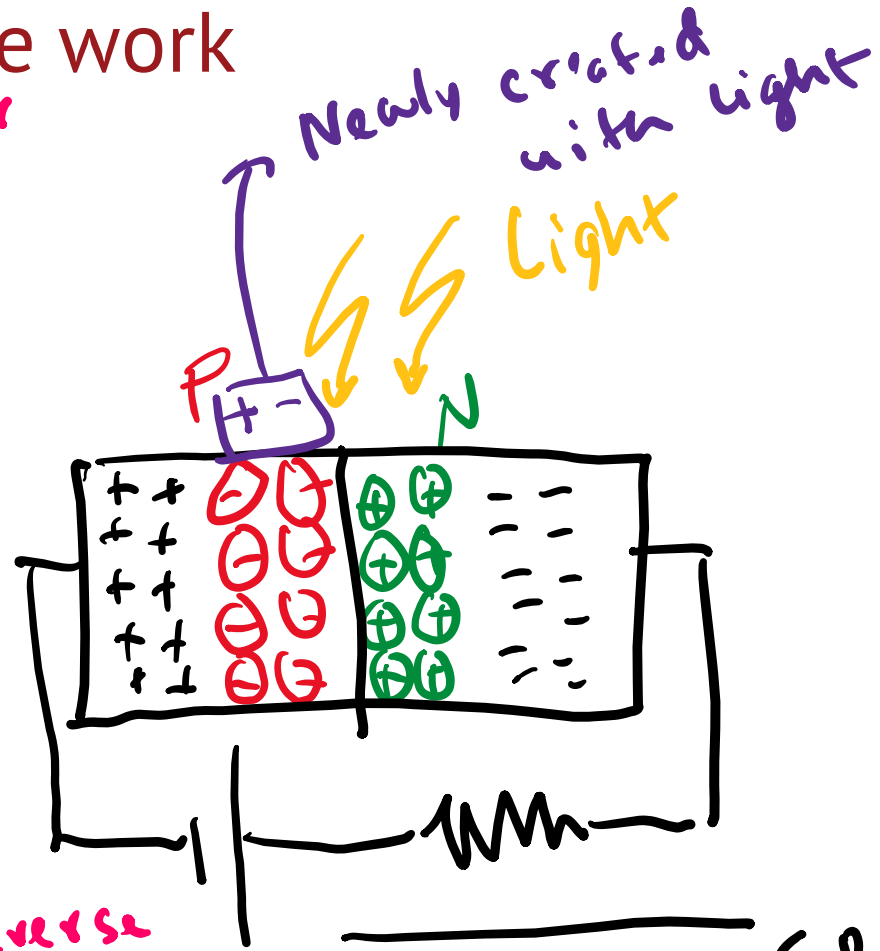
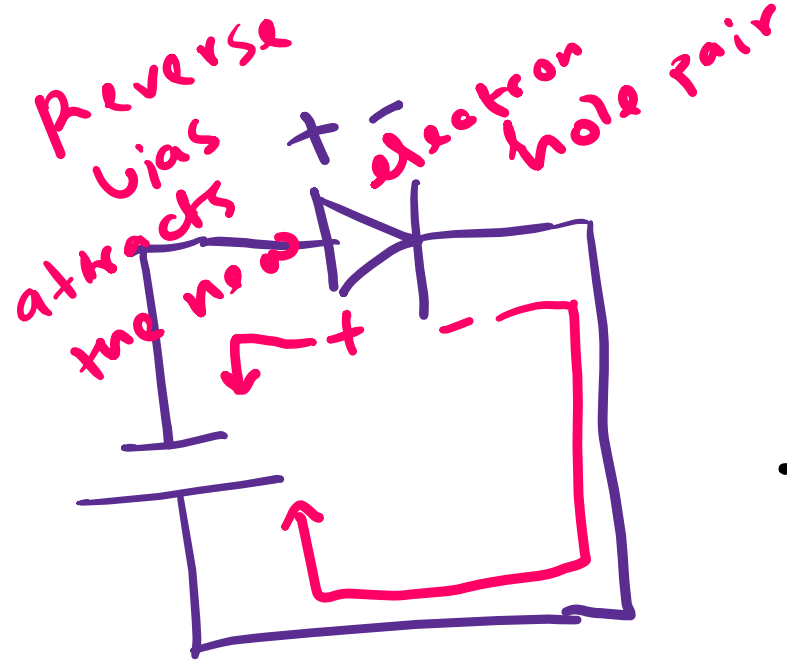
LED
emit red
light



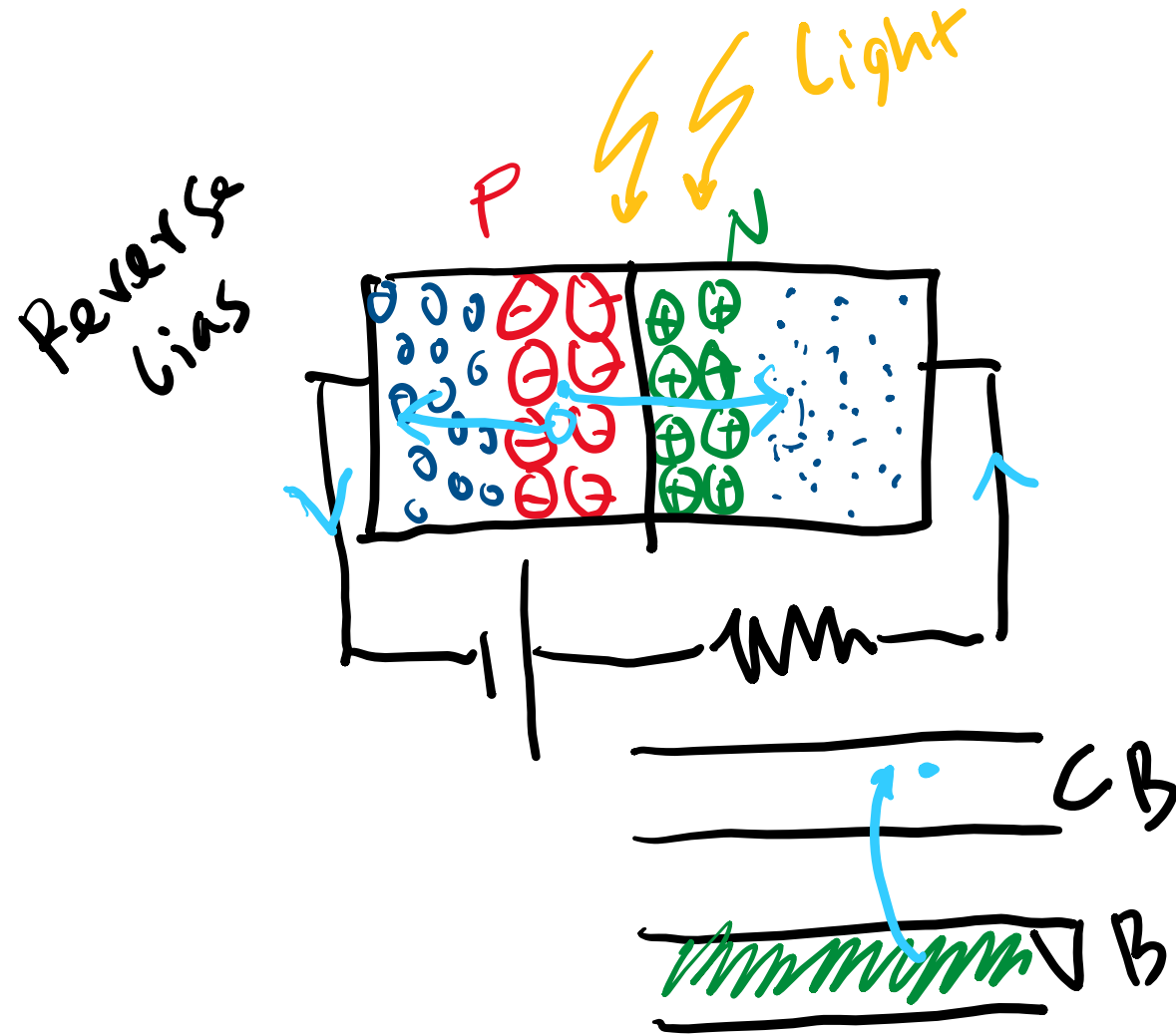
PD captures
blue and NIR
light

PD won't
capture the
red
light.

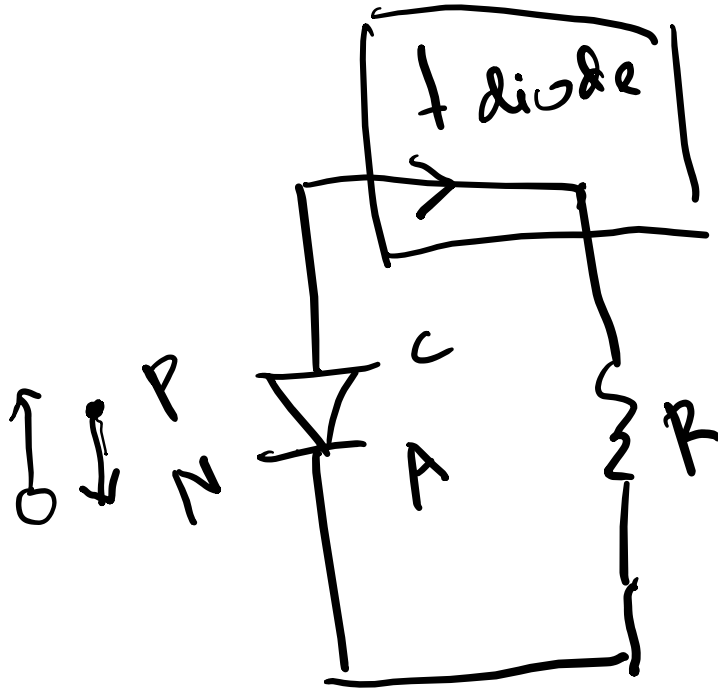
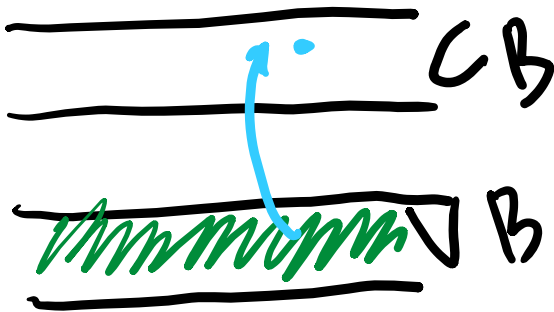
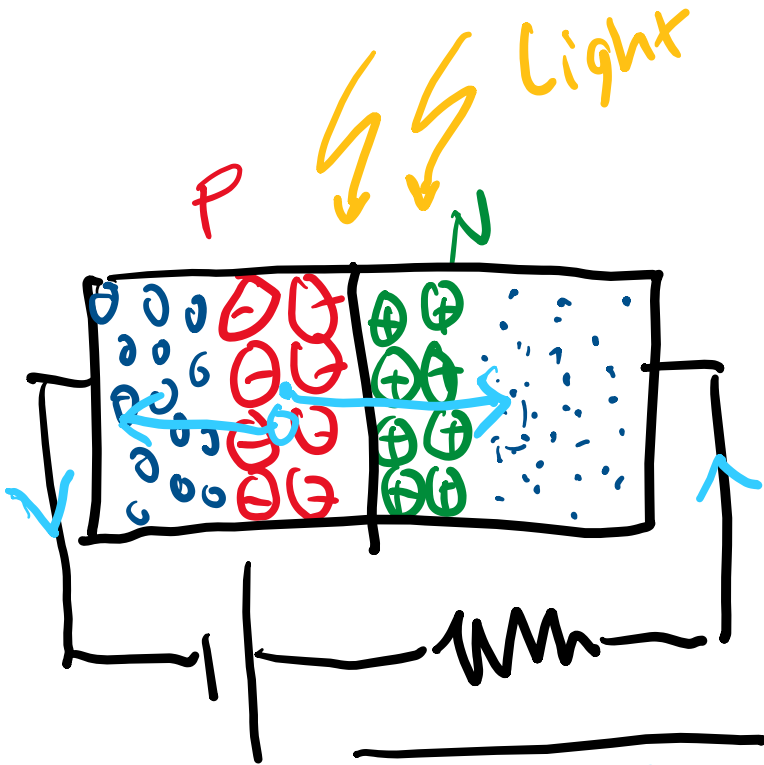
How does a photodiode work



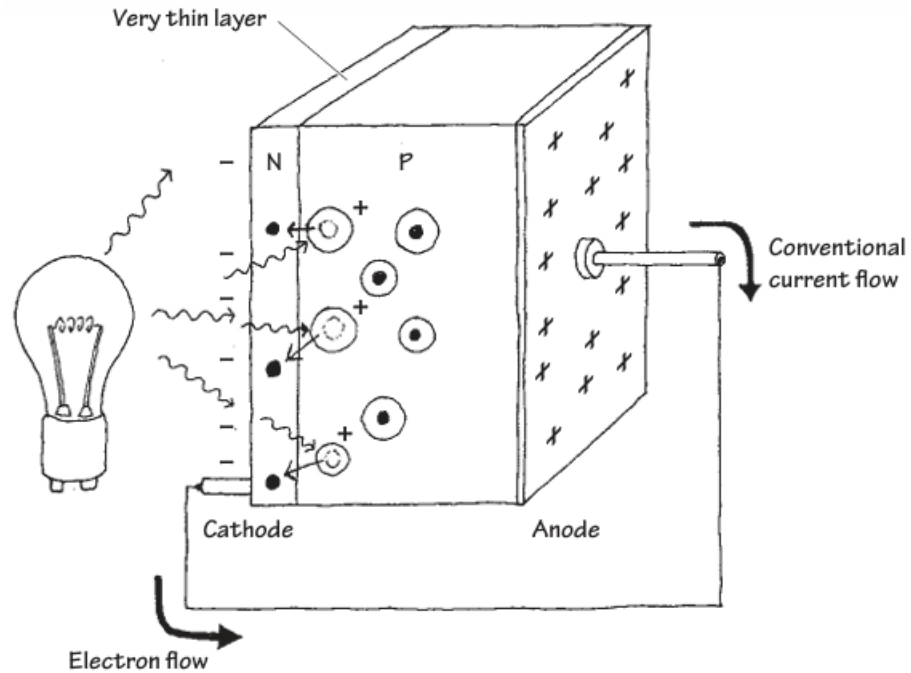
How does a photodiode work



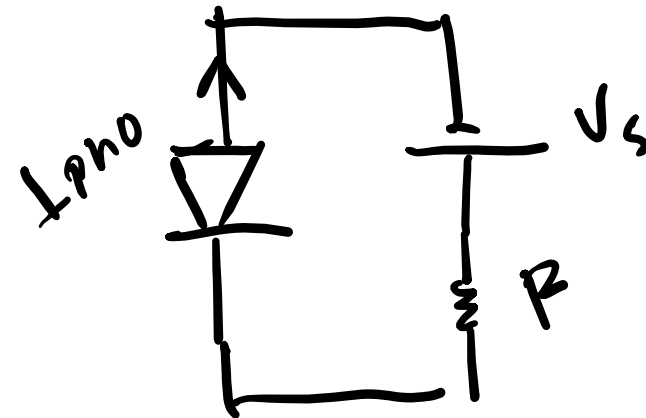
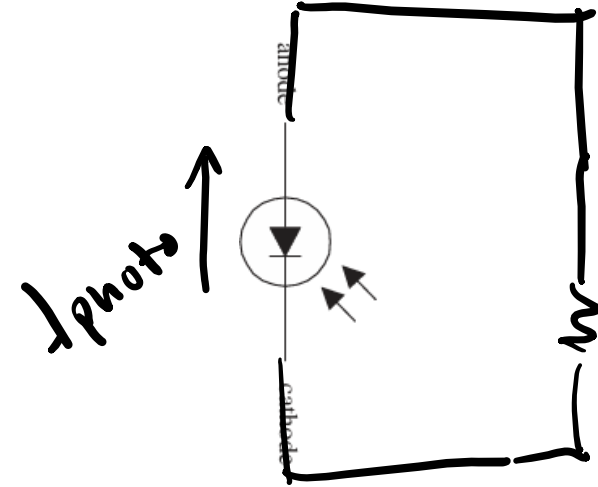
How does a photodiode work



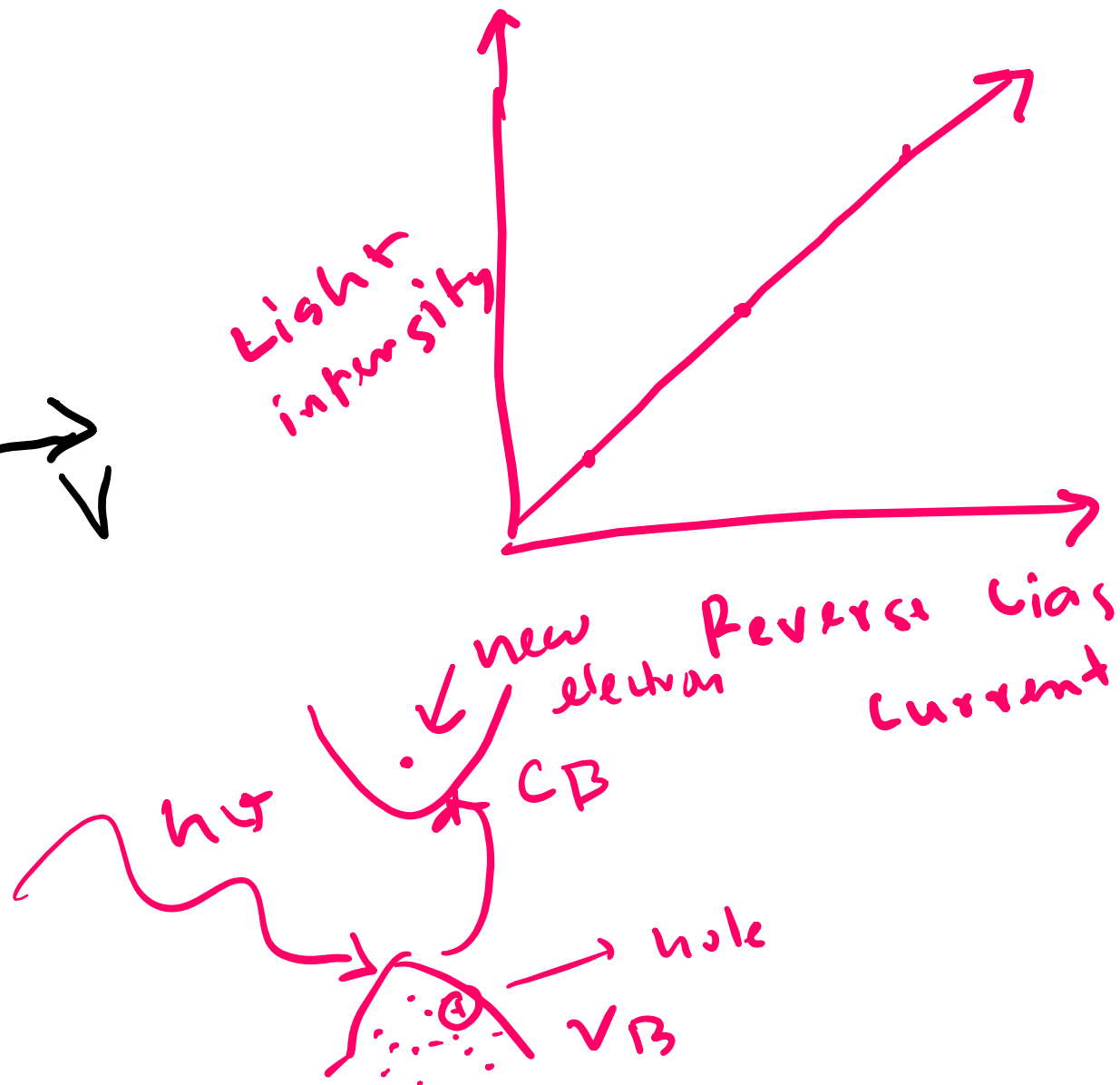
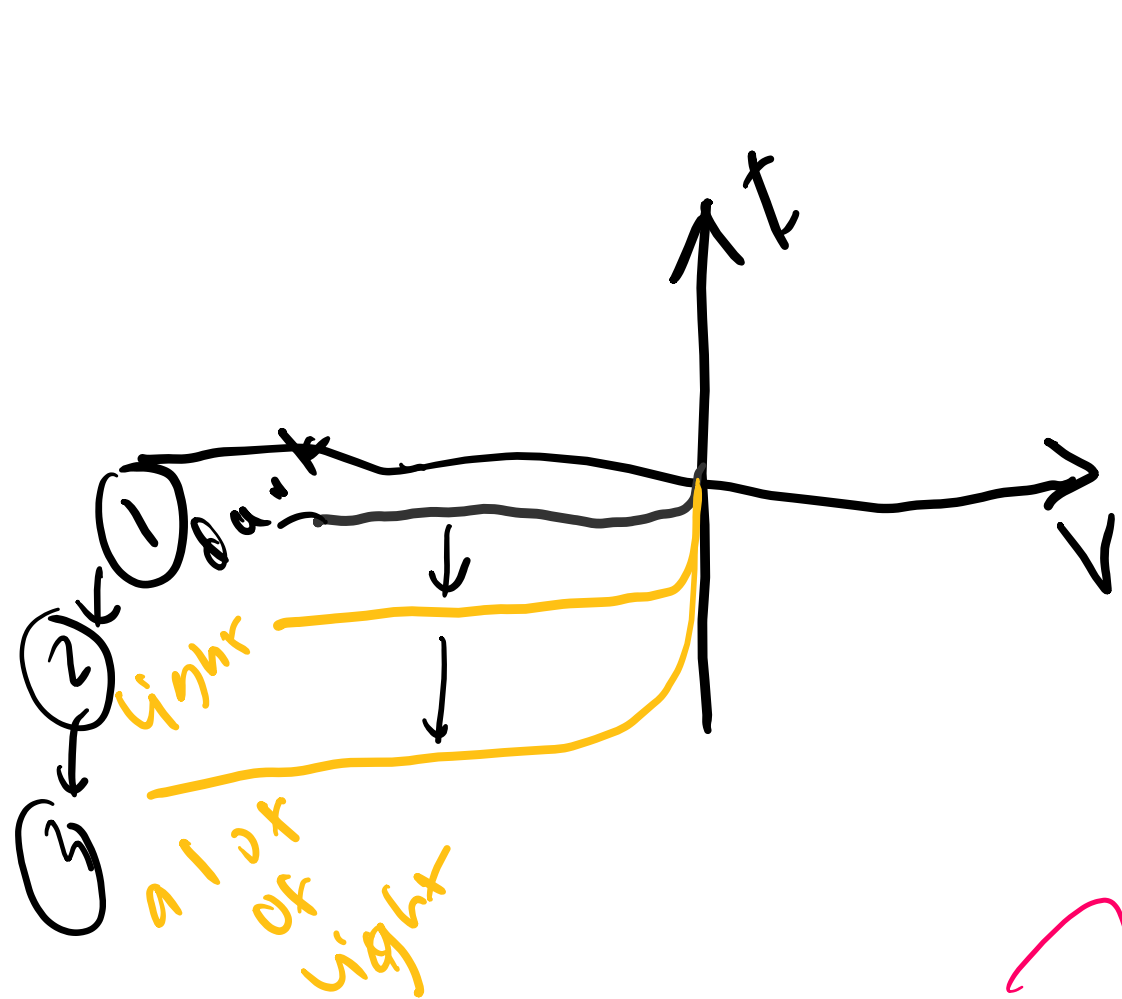
Photodiode



Reverse bias



IV of photodiode



The PD in our demo board

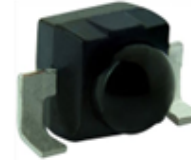


www.vishay.com

VEMD2023SLX01

Vishay Semiconductors

Silicon PIN Photodiode



DESCRIPTION

VEMD2023SLX01 is a high speed and high sensitive PIN photodiode in a miniature side looking, surface mount package (SMD) with dome lens and daylight blocking filter. Filter is matched with IR emitters operating at wavelength of 830 nm to 950 nm. The photo sensitive area of the chip is 0.23 mm².

FEATURES

- Package type: surface mount
- Package form: side view
- Dimensions (L x W x H in mm): 2.3 x 2.55 x 2.3
- AEC-Q101 qualified
- High radiant sensitivity
- Daylight blocking filter matched with 830 nm to 950 nm IR emitters
- Fast response times
- Angle of half sensitivity: $\varphi = \pm 35^\circ$
- Package matched with IR emitter series VSMB2943SLX01
- Floor life: 4 weeks, MSL 2a, acc. J-STD-020
- Lead (Pb)-free reflow soldering
- Material categorization: For definitions of compliance please see www.vishay.com/doc?99912



APPLICATIONS

- High speed photo detector
- Infrared remote control
- Infrared data transmission
- Photo interrupters
- IR touch panels

PRODUCT SUMMARY

COMPONENT	I_{rs} (μA)	φ (deg)	$\lambda_{0.5}$ (nm)
VEMD2023SLX01	10	± 35	750 to 1050

Note

- Test conditions see table "Basic Characteristics"

ORDERING INFORMATION

ORDERING CODE	PACKAGING	REMARKS	PACKAGE FORM
VEMD2023SLX01	Tape and reel	MOQ: 3000 pcs, 3000 pcs/reel	Side view

Note

- MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25^\circ\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Reverse voltage		V_R	60	V
Power dissipation	$T_{amb} \leq 25^\circ\text{C}$	P_V	215	mW
Junction temperature		T_J	100	$^\circ\text{C}$
Operating temperature range		T_{amb}	-40 to +100	$^\circ\text{C}$
Storage temperature range		T_{stg}	-40 to +100	$^\circ\text{C}$
Soldering temperature	Acc. reflow solder profile fig. 7	T_{sd}	260	$^\circ\text{C}$
Thermal resistance junction/ambient	Acc. J-STD-051	R_{thJA}	250	K/W

<https://www.vishay.com/docs/83493/vemd2023slx01.pdf>

PD characteristics

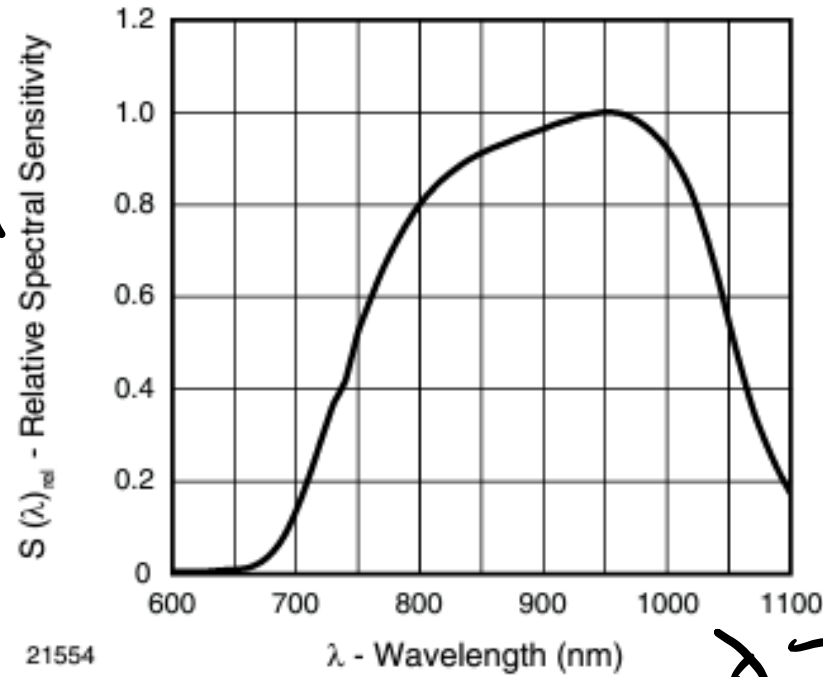


Fig. 5 - Relative Spectral Sensitivity vs. Wavelength

10^{-9}
nA

10^{-6}
nA

BASIC CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

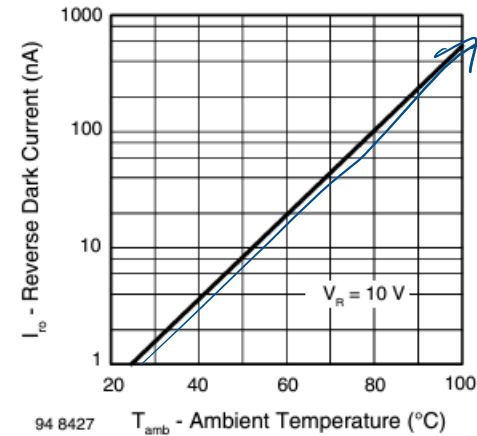


Fig. 1 - Reverse Dark Current vs. Ambient Temperature

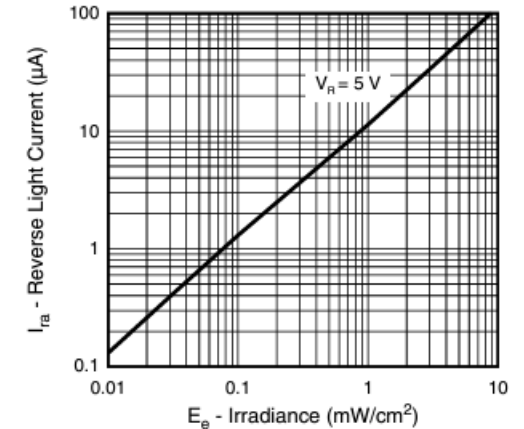
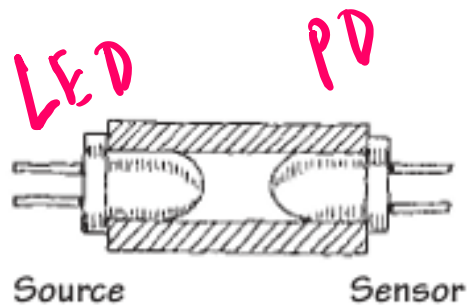
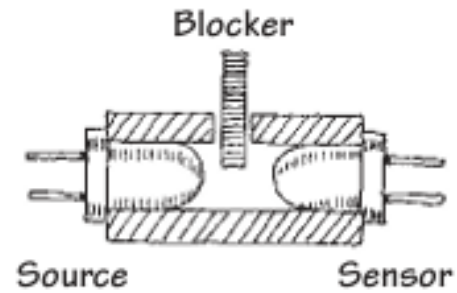


Fig. 3 - Reverse Light Current vs. Irradiance

(a) Closed pair



(b) Slotted coupler



(c) Reflective pair

